

Diabetic Retinopathy Classification Using Deep Learning Techniques



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Spring 2026

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ACKNOWLEDGEMENTS

We have completed this research work with the help of ALLAH Almighty who always helped us whenever we were in trouble. We would like to acknowledge our Internal Supervisor Dr. Hussam Ali for all his diligence, guidance and supervision that enabled us to complete this research work. We express our sincere and heartfelt gratitude to our External Supervisor, Dr. Muhammad Fayyaz (Head of Computer Science Department, FAST NUCES, CFD Campus), for his continuous motivation, inspiring guidance, and valuable research insights. We are truly thankful for his constant support and belief in our abilities throughout this journey. We want to thank our Project Manager Prof. Dr. Muhammad Ilyas (Dean of computing faculty, University of Sargodha) for his continuous supervision and guidance during the whole period of our research. We are equally indebted to our parents, friends and siblings who always helped and encouraged us whenever we lost heart during this tedious and laborious work.

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LIST OF ABBREVIATIONS

DL	Deep Learning
DR	Diabetic Retinopathy
NPDR	Non-proliferative diabetic retinopathy (NPDR)
PDR	Proliferative diabetic retinopathy
CNN	Convolutional Neural Network
ML	Machine Learning
GAP	Global Average Pooling
CLEHE	Contrast Limited Adaptive Histogram Equalization
ReLU	Rectified Linear Unit
APSO	Adaptive Particle Swarm Optimization
ViTs	Vision Transformers
SATT	Selective Attention
GATT	Global Attention
EN-B4	EfficientNet-B4
RGB	Red, Green, Blue
SE	Squeeze-and-Excitation
MBConv	Mobile Inverted Bottleneck Convolution
MSAB	Multi Stage Attention Block
SLR	Systematic Literature Review
FC	Fully Connected
QWK	Quadratic Weighted Kappa
DenseNet-201	DN-201

ABSTRACT

Diabetic retinopathy (DR) has become a leading cause of blindness worldwide and the manual screening of color fundus images to detect DR at early stages is expensive and time consuming. To address this problem, we proposed a CNN model with attention mechanism for automated DR classification using retinal fundus images. The proposed method focuses on improving classification performance by enhancing image quality through preprocessing techniques such as normalization and data augmentation. EfficientNet-B4 and DenseNet-201 are used as backbones for feature extraction. To further strengthen the model, a Multi-Stage Attention (MSAB) Block consisting of Global Attention (GATT) and Selective Attention (SATT) modules is introduced to highlight important regions of the retina. These two attention modules are used for spatial and channel-wise feature refinement. The model is trained and tested on two benchmark datasets APTOS and IDRiD. Experimental results show that the proposed framework, yielded 81.6% and 82.2% accuracy, 80.53% and 81.96% F1-score, 86.57% and 90.21% QWK scores on the APTOS and IDRiD datasets using EfficientNet-B4 as backbone, respectively. Similarly, by using DenseNet-201 as backbone the proposed framework achieved 81.4% and 94.76% accuracy 80.31% and 94.69% F1-score, 84.81% and 98.34 % QWK scores on the APTOS and IDRiD datasets, respectably. This shows that the proposed approach achieves competitive performance that is at par with recently reported works on DR classification.

Chapter 1

Introduction

1.1 Introduction

The human eye is an amazing sensory organ that is the gateway to visual perception. It detects light and sends signals to the brain through the optic nerve to form images, colors and depth perception [1]. This complex process allows us to recognize face's help detect objects and do many activities that require visual input.

But many medical conditions can impair the eye and cause significant visual loss, Diabetic Retinopathy (DR) is one of them. DR is a complication of diabetes that leads to vision loss. It happens due to prolonged diabetes that damage blood vessels in the retina and is the prominent reason for blindness worldwide. It affects the thin blood vessels in the retina that are responsible for capturing light and can cause blockage and bleeding in the retina, and ultimately leads to permanent vision loss if not detected early.

Roughly, one-third of the 463 million people worldwide who have diabetes are affected by it [2]. The International Diabetes Federation (IDF) projects that there will be a significant increase in the global diabetes population from roughly 552 million in 2035 to 642 million in 2040 [3]. DR can be classified into five distinct stages including No DR(DR0), Mild (DR1), Moderate (DR2), Severe Non-proliferative (DR3), and Proliferative (DR4) [4]. Non-proliferative diabetic retinopathy (NPDR) causes retinal swelling and leakage of tiny blood vessels while Proliferative diabetic retinopathy (PDR) is the most severe stage of the disease, in which new blood vessels start developing in the retina.

Deep learning (DL) methods have shown great promise in automatically detecting the accurate stages of DR. These models showed great accuracy in detecting DR but still many ophthalmologists heavily rely on manual detection. Manual detection of DR is time-consuming and requires trained clinical experts to analyze digital color fundus images. However, the delayed outcomes can result in a lack of follow-up and severe outcomes for patients [5]. Convolutional

Neural Networks (CNNs) have showed great performance in recognizing DR-related features, such as microaneurysms, hemorrhages, and exudates, directly from images without the need for handcrafted features [6]. Despite the progress in machine learning based DR detection, we need to address the current limitations. Existing models although work well in many controlled settings, fail to generalize across different datasets and real-world clinical environments.

Many DL techniques showed great accuracy in detecting and classifying DR but now the focus is towards using Hybrid techniques which combines different CNN models to make the output more clinically explainable because they don't only classify DR but also provide ophthalmologists enough information to use them in real time diagnosis [7]. Moreover, hybrid models also help's in detecting progression of disease.

This research will focus on advanced computational techniques, state of the art DL frameworks and robust validation processes for DR classification. Our model will integrate CNNs for local feature extraction with multi-stage attention mechanism for highlighting critical regions, and employs feature fusion to enhance the separation between classes. The goal is to handle class imbalance in DR datasets through architecture-level improvements and to validate the model on benchmark datasets like APTOS and Eye PACS.

The main goal of this research is to develop robust AI-based solution that is mostly required in both urban hospitals and rural healthcare centers [8]. This solution will not only reduce the workload of specialists but also make early classification more accurately. Moreover, with timely intervention, this technology could play a vital role in preventing vision loss and improving the quality of life for millions of diabetic patients worldwide.

1.2 Research Challenges

Several research challenges exist in DR classification such as class imbalance, limited dataset diversity and the difficulty of distinguishing subtle differences between adjacent DR stages. These challenges significantly impact classification and model generalization. A detailed discussion on these challenges is provided in the following subsections.

1.2.1 Class Imbalance

Class imbalance is a major challenge in DR classification, where no DR and mild stage DR images are significantly more in size than advanced stages [9]. This uneven distribution biases model towards majority class, resulting in poor sensitivity for severe DR cases. The scarcity of advanced DR images limits the model's ability to learn representative features resulting in reduced classification accuracy. Addressing class imbalance is therefore essential for reliable and clinically meaningful DR classification.

1.2.2 Limited dataset diversity

Limited dataset diversity remains a critical challenge in DR research. Many publicly available datasets fail to capture variations in image resolution, illumination, contrast, and color distribution encountered in real-world clinical environments. This limited variability increases the risk of overfitting and reduces model performance when applied to unseen clinical data. Therefore, working on dataset diversity is essential for developing reliable, generalizable, and clinically applicable DR classification systems.

1.2.3 Subtle Differences between DR stages

Distinguishing between adjacent DR stages remains challenging due to subtle visual differences in retinal lesions. Early and intermediate stages often share overlapping lesion patterns such as microaneurysms and mild hemorrhages. These fine grinded variations are indistinguishable to learning models. Accurate discrimination of subtle lesion progression is essential for precise DR classification.

1.2.4 Model Interpretability:

Model interpretability remains a critical challenge in DR classification. Deep learning models are often viewed as “black boxes”, limiting their use in clinical environment. Although they provide accurate predictions but without clear explanations reduces clinical trust and limits acceptance in medical practice. Without interpretable insights, it becomes difficult to validate model predictions and ensure reliable clinical deployment.

1.3 Research Questions

RQ1: Can a deep CNN model with attention mechanism outperform standard CNN models in accurately classifying different stages of diabetic retinopathy? [Discussed in chapter 4].

RQ2: How does the integration of attention mechanisms help deep learning models handle image quality variations, class imbalance, and improve generalization across diverse real-world DR datasets? [Discussed in chapter 4].

1.4 Research Contributions

The contributions are summarized below:

- Developed a hybrid deep learning architecture with multi-stage attention block for DR classification using retinal fundus images.
- Applied image preprocessing techniques such as normalization, enhancement, and data augmentation to improve image quality and handle class imbalance.
- Integrated an attention mechanism to focus on important retinal regions, enhancing feature extraction and classification performance.
- Evaluated the model using standard performance metrics, including accuracy, precision, recall, F1-score, and QWK.

1.5 FYP Document Organization

This FYP document is organized into five chapters. Figure 1.1 illustrates the organization and a chapter-wise description is given below:

Chapter 1 presents a detailed introduction regarding DR classification. In this regard, background of the disease is briefly discussed. Moreover, research challenges, gaps and objectives of the proposed study are also presented.

Chapter 2 gives a review of state-of-the-art techniques for DR classification using traditional, deep learning, hybrid and vision transformer approaches. Additionally, this section presented the detailed systematic literature review (SLR).

Chapter 3 describes the proposed methodology for DR classification including dataset preparation, feature extraction using pre-trained CNN models, and the proposed Multi stage attention block implementation details.

Chapter 4 presents the experimental results for proposed DR classification framework. This chapter includes a detailed discussion regarding evaluation matrices, comprehensive comparison with state-of-the-art studies and detailed ablation experiments.

Chapter 5 concludes the proposed work and also discusses limitations and provides directions for future research.

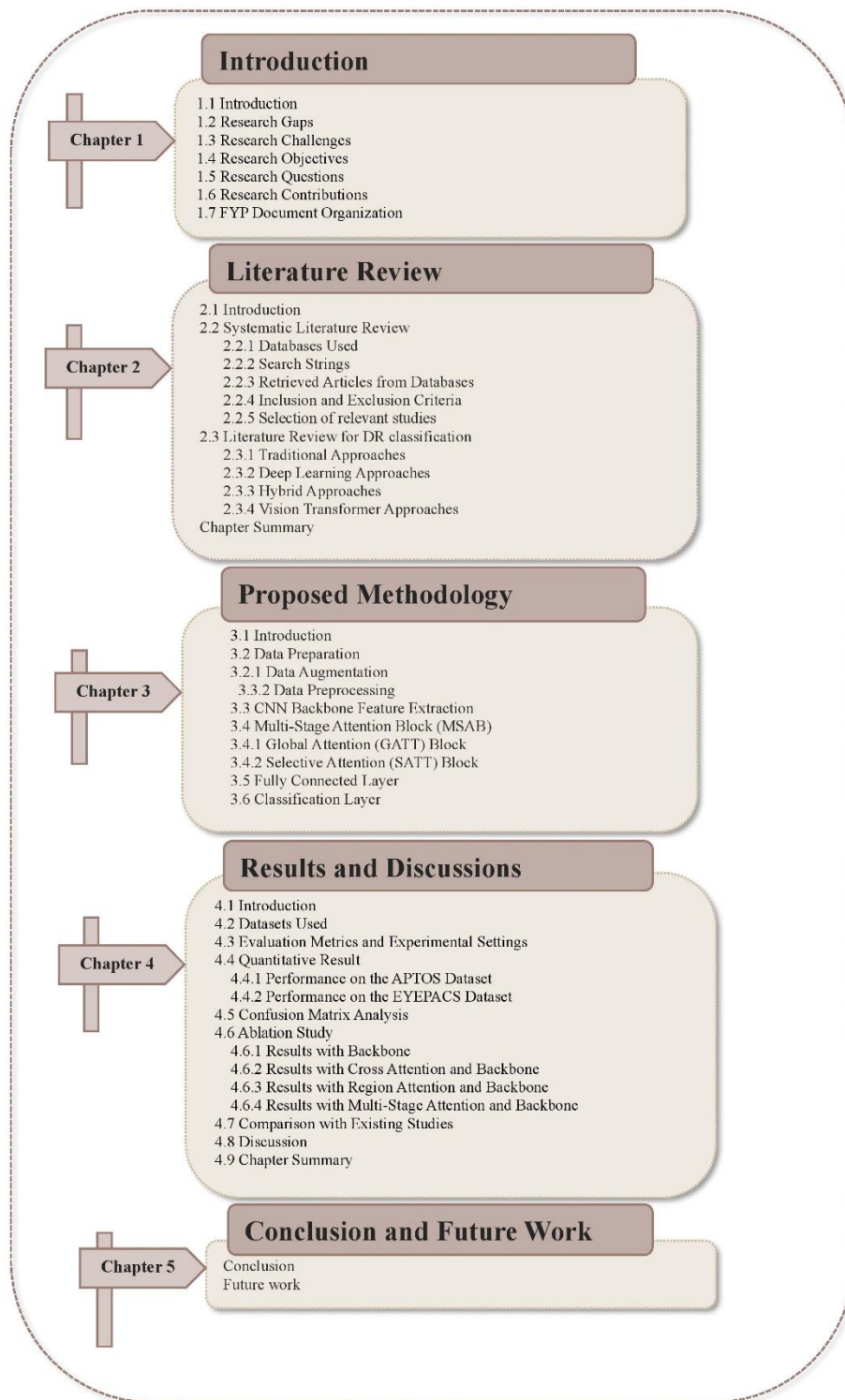


Figure 1.1: Detail overview of Document

Chapter 2

Literature Review

2.1 Introduction

This chapter presents a comprehensive and systematic review of existing research related to DR classification using computational intelligence techniques. The primary objective of this chapter is to analyze state-of-the-art methods, identify their strengths and limitations, and highlight unresolved challenges in automated DR classification. The literature is categorized into traditional machine learning and deep learning-based methods, hybrid deep learning frameworks, and vision transformer-based models. Special attention is given to recent studies (2020–2025) that focus on multi-class DR classification, attention mechanisms, feature fusion, interpretability, and dataset diversity. This critical review provides a strong foundation for motivating the proposed multi-stage attention-based hybrid methodology presented in Chapter 3.

2.2 Systematic Literature Review

To ensure a structured and transparent selection of relevant studies, a Systematic Literature Review (SLR) was conducted following the PRISMA guidelines. The SLR approach was adopted to comprehensively analyze recent advancements in DR classification while minimizing selection bias and redundancy.

2.2.1 Databases Used

Multiple well-established scientific databases were explored to retrieve high-quality and peer-reviewed research articles. These include IEEE Xplore, PubMed, SpringerLink, ScienceDirect, and Google Scholar. The use of multiple databases allowed broad coverage of research across medical imaging, artificial intelligence, and ophthalmology. The links are provided in Table 2.1.

Table 2.1: Databases Used for Searching

Databases Used	Links
IEEE Xplore	https://ieeexplore.ieee.org/
PubMed	https://pubmed.ncbi.nlm.nih.gov/
SpringerLink	https://link.springer.com/
ScienceDirect	https://www.sciencedirect.com/
Google Scholar	https://scholar.google.com/

2.2.2 Search Strings

Carefully designed search strings were applied to retrieve a wide range of relevant studies focusing on deep learning and hybrid approaches for DR grading. These search strings combined key terms such as diabetic retinopathy, CNN, transfer learning, hybrid deep learning, and attention mechanisms. The use of Boolean operators (“AND” and “OR”) in the search strategy improved search precision while maintaining comprehensive coverage. The searching strings are provided in Table 2.2.

Table 2.2: Search Strings Used to retrieve results

Search string ID	Search strings
1	(“diabetic retinopathy classification” AND “transfer learning” OR “CNN”)
2	(“diabetic retinopathy” AND “hybrid deep learning”)
3	(“diabetic retinopathy” AND “attention-based hybrid deep learning”)

2.2.3 Retrieved Articles from Databases

Predefined search strings defined in Table 2.2 for the retrieval of articles from multiple databases, including IEEE Xplore, PubMed, SpringerLink, ScienceDirect and Google Scholar. The search retrieved a total of 308 articles as shown in the figure 2.1. ScienceDirect has the highest number of records (95), followed by Google Scholar (90). SpringNature retrieved 69 records, while IEEE Xplore yielded 46 records. PubMed with the fewest articles, retrieved only 8 records. Overall, the initial search retrieved many relevant studies related to diabetic retinopathy classification using deep learning and hybrid techniques, which were then filtered for duplicate removal, screening and final study selection as part of the systematic review process.

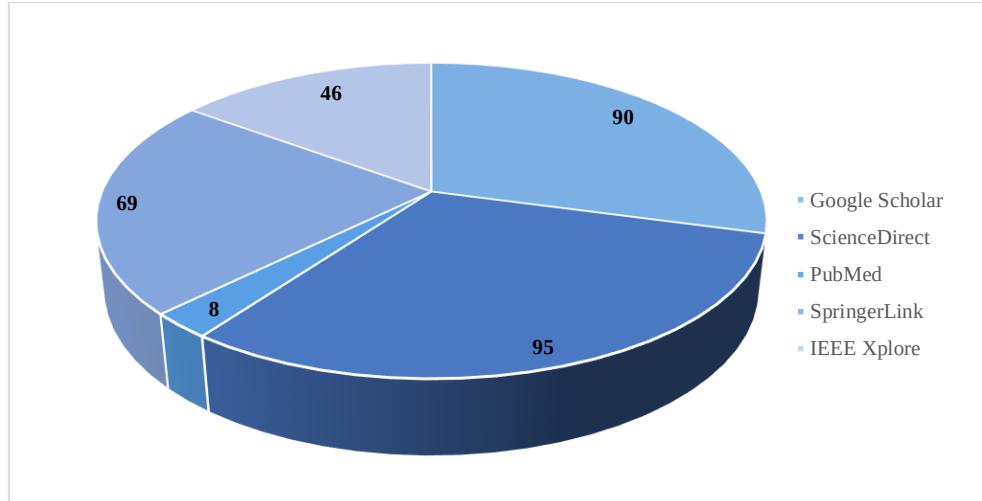


Figure 2.1: Number of Retrieved Articles from Databases

2.2.4 Inclusion and Exclusion Criteria

To maintain relevance and quality, strict inclusion and exclusion criteria were applied. Only studies focusing on multi-class DR classification using deep learning or hybrid approaches and employing publicly available datasets were included. Articles published before 2020, non-English papers, review-only studies, and works relying solely on traditional machine learning were excluded. This filtering ensured alignment with the objectives of the proposed research.

Table 2.3: Inclusion and Exclusion Criteria

Criteria	Inclusion Criteria	Exclusion Criteria
Scope of Study	Studies mainly focusing on Multi Class diabetic retinopathy (DR) classification.	Studies not related to diabetic retinopathy classification.
Techniques Used	Studies implementing hybrid deep learning techniques (e.g., CNN with attention, feature fusion, transformers, or vision transformer).	Research using only traditional machine learning techniques without hybrid or deep learning models.
Datasets Used	Research mostly done on well-known fundus images datasets	Studies that do not use publicly available datasets.
Publication Type	Journal articles, conference papers, or high-impact open access publications.	Reports, survey papers, review articles, theses, posters
Publication Period	Articles published between 2020–2025.	Articles published before 2020
Language Used	Studies written in the English language.	Studies not written in the English language.

2.2.5 Selection of relevant studies

This review follows the PRISMA guidelines for searching and selection of articles [10]. A total of 308 related papers published between 2020 to 2025 were retrieved from the databases listed in Table 2.1. After removing 30 duplicate records using the reference management software Endnote, 278 unique studies were kept for screening. During the title and abstract screening phase, 200 articles were excluded for not meeting the inclusion criteria. This screening narrowed the selection to 78 full-text articles for detailed evaluation. Following the full-text review, 40 articles were excluded due to unsuitable datasets, absence of hybrid deep learning techniques or lack of focus on DR classification. After full text review, 38 articles were retained but 8 were excluded after further evaluation. Finally, 30 high-quality studies meeting the inclusion criteria and strong alignment with the research objectives were selected. The flow diagram of the article searching and selection is shown in Figure 2.

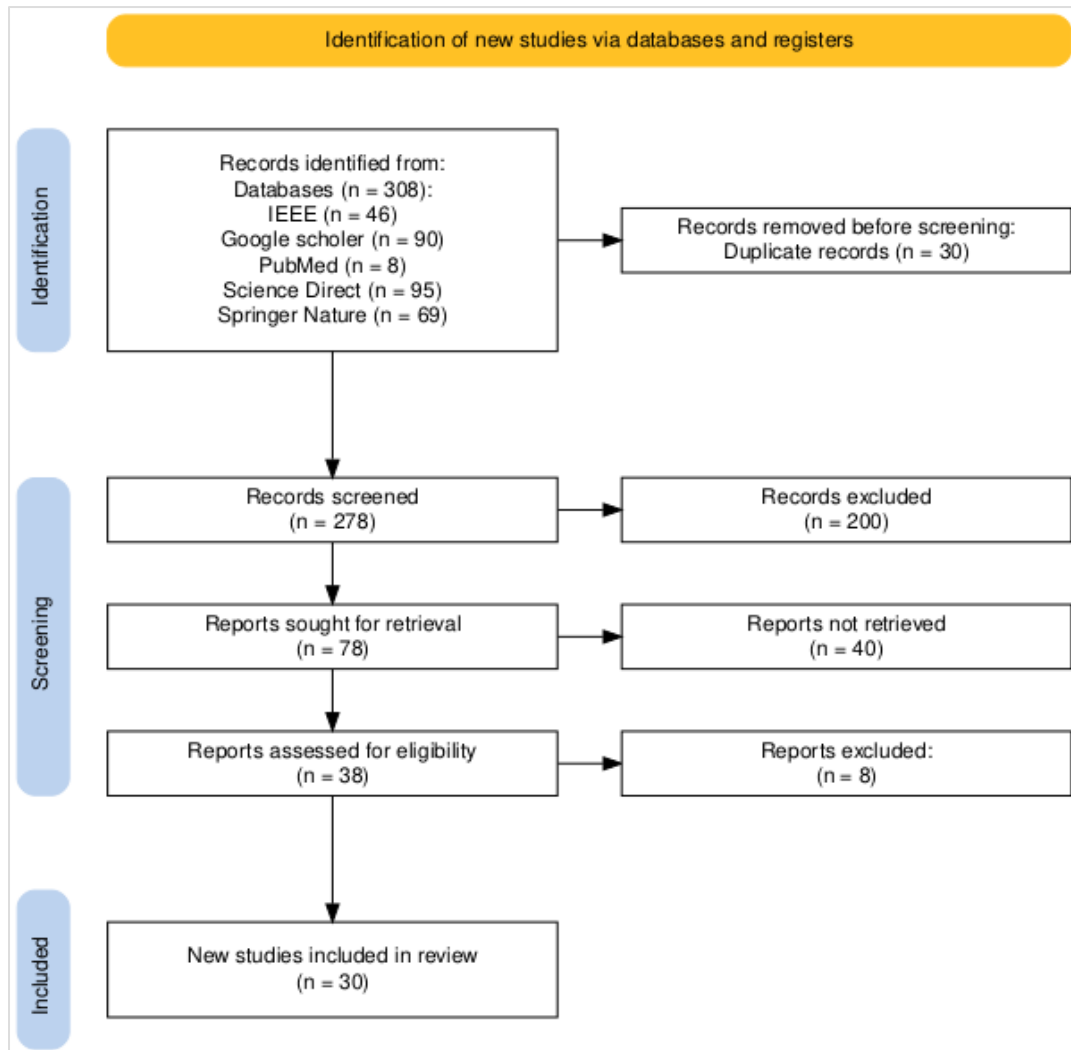


Figure 2.2: PRISMA Flow Diagram

2.3 Literature Review for DR classification

Automated DR classification has evolved significantly over the past decade. Early approaches relied heavily on handcrafted features and traditional classifiers, while recent advancements focus on deep learning, hybrid, attention mechanisms, and transformer-based architectures. The following sections critically examine these methodologies.

2.3.1 Traditional Approaches

Traditional DR detection methods rely on handcrafted features with traditional machine learning classifiers. According to relevant literature, mostly techniques initially applied image processing techniques such as contrast enhancement, morphological filtering, and edge detection to analyze retinal structures [11]. Moreover, these techniques focus on key features that include blood vessel segmentation, optic disc localization, and lesion-specific textures (microaneurysms and hemorrhages) including classifiers such as SVM, Random Forest, k-NN, and Decision Trees are applied to grade DR severity and classification [12]. Although traditional approaches are computationally efficient and interpretable but highly sensitive to image quality and illumination variations.

The study by [7] used image enhancement and manual features extraction achieved about 65-80% accuracy. While global DR screening studies using logistic regression and classical ML on Messidor and clinical datasets reported 70-80% performance [8]. In another study [13] authors proposed 16 class classification method using Support Vector Machine (SVM) classifier capable of predicting individual and multi-lesion abnormalities. The proposed pipeline integrated Gaussian mixture model (GMM), K-means, Maximum a Posteriori (MAP) algorithm, Principal Component Analysis (PCA), Grey level co-occurrence matrix (GLCM), and SVM for disease diagnosis. The proposed method reported an accuracy of 77.3% on DIARETDB1 dataset. Due to their poor scalability and heavy dependence on manual feature design, these traditional approaches have been replaced with deep learning-based approaches. The summary of traditional approaches is presented in Table 2.4.

Table 2.4: Traditional Approaches for Diabetic Retinopathy

Ref	Year	Method/Approaches	Dataset	Acc (%)
[7]	2015	Image Enhancement, Manual Feature Extraction	Asian hospital datasets	65–78%
[8]	2016	Screening Tools, Logistic Regression, Classical ML	Messidor, Duke, Clinical datasets	70–80%
[13]	2022	16-class SVM-based DR classification	DIARETDB1	77.3%

2.3.2 Deep Learning Approaches

Recently, applications of deep learning models showed promising performances in all areas of research. In deep learning-based methods, CNNs learn meaningful and complicated features (microaneurysms, hemorrhages, exudates) directly from retinal images by eliminating the limits of manual, handcrafted features. Popular models like VGG, Res-Net, Dense-Net, and Inception consistently outperform traditional methods in multi-class DR grading. DL models work especially well on large datasets such as EyePACS and APTOS, where thousands of images help them learn strong, generalizable features for DR classification.

Moreover, recent reviews indicate that integrating multiple deep learning models enhances feature representation and leads to improved diagnostic accuracy in DR classification tasks [14, 15]. For instance, CNN-based systems are often paired with preprocessing steps, normalization, resizing, and lesion enhancement and use approaches like transfer learning, fine-tuning, and data augmentation to further boost accuracy when data is limited [16]. Several studies in this domain have explored enhanced CNN pipelines to improve accuracy, robustness, and generalization.

Another deep learning approach [17] used VGG19 and InceptionV3 models to trained on two segmentation-based features such as color-based Blood Vessel (BV) segmentation and color-based Contrast-Limited-Adaptive-Histogram-Equalization-Top Hat (CLAH-TH) segmentation. Moreover, the authors applied two strategies such as feature fusion and data augmentation in proposed work to enhance the DR classification results. The authors reported an accuracy of 96% on the Kaggle DR Detection dataset. In another study [18] a deep learning-based Retinopathy Severity

Grading Network (RSG-NET) is proposed to classify DR for four and two stages. The authors applied extensive preprocessing and data augmentation techniques to enhance fundus image representation and handle class imbalance issues, respectably. The proposed RSG-NET evaluated on the Messidor-1 dataset and achieved accuracies of 99.36% and 99.37% for four stage and two stage, respectably.

To achieve highest possible accuracy in lowest possible computational time, authors merge three DR datasets APTOS, Messidor2, IDRiD and applied various preprocessing and three augmentation (geometric, photometric, and elastic deformation) techniques to enhance image quality and class imbalance respectably. Authors developed RetNet-10 by optimizing a base CNN through careful hyperparameter tuning. The proposed RetNet-10 model outperformed six state-of-the-art architectures and achieved highest performance accuracy of 98.65% [19]. In another study authors presented a two-stage transfer learning method which fine tunes early and higher layers of pre-trained CNN to extract domain specific and discriminative features respectably. The proposed approach applied PCA and gradient boosting for feature reduction and for final classification. Evaluated on two challenging datasets EyePACS and Messidor model achieved an accuracy of 99.73% and 98.88 [20].

Similarly, in [21] authors trained three distinct CNN based models for binary classification, early detection, and severity grading with three convolutional layers, a global average pooling layer, and three fully connected layers. Evaluated with five-fold cross-validation, the proposed method achieved 96.2% accuracy for binary classification, 86% for early detection, and 79.4% for severity prediction.

Most recently, efforts have been made to improve model interpretability by integrating Explainable Artificial Intelligence (XAI) techniques with deep learning models. In [22] authors employed various transfer learning, machine learning and neural network models along with a custom CNN architecture optimized using softmax and categorical cross-entropy for binary and multi-class classification. Evaluated on APTOS dataset the proposed model achieved an accuracy of 95.27% along with visual explanations to support clinical decision-making. In another study [6] a knowledge distillation-based framework is proposed, where a fusion-based teacher model combined ResNet152V2 and Swin Transformer transfers learned knowledge to a lightweight student model build using Xception architecture with a customized Convolutional Block Attention Module (CBAM) to learn rich and discriminative feature representations. Authors applied four-

stage image enhancement pipeline for enhancing image quality. Evaluated on two benchmark datasets (APTOS and IDRiD) the proposed method achieved 100% and 99.04% accuracy on APTOS and 98.05% and 94.17% on IDRiD dataset for binary and multi-class classification respectively.

The study by [23] developed three approaches for early DR prediction by using DenseNet-121 and AlexNet with Principal Component Analysis (PCA) for feature extraction. Three proposed approaches utilized low-dimensional features, important and low-dimensional features before and after PCA and ANN with the radiomic features of Dense-121 and AlexNet. The study reported that radiomic features formed by combining CNN-based (DenseNet-121 and AlexNet) with the handcrafted features (Wavelet Transform (DWT), Local Binary Pattern (LBP), Fuzzy colour histogram (FCH), and Gray Level Co-occurrence Matrix (GLCM)) methods achieved an accuracy of 99.56%. In [24] hybrid VGG16-XGBoost and DenseNet-121 architectures are proposed for DR detection and classification. Authors applied various preprocessing and data balancing techniques for image enhancement and class imbalance. The proposed work reported that hybrid and DenseNet-121 network achieved an accuracy of 79.50% and 97.30% respectively. Although DL models give higher accuracy, robustness, and generalization but still struggle with issues like class imbalance and high training costs. The summary of existing hybrid approaches for DR is presented in Table 2.5.

Table 2.5: Deep Learning Approaches for Diabetic Retinopathy (2023-2025)

Ref	Year	Method/Approaches	Dataset	Acc (%)
[18]	2025	Retinopathy Severity Grading Network (RSG-NET) with extensive preprocessing and augmentation techniques.	Messidor-1	99.36%
[21]		Lightweight CNN + CLAHE + Augmentation + GAP	APTOS	97.5%
[17]	2024	BV segmentation + CLAHE-TH enhancement, fused features into VGG19 / InceptionV3 classifier.	Kaggle DR detection dataset	96%
[22]		Custom CNN with convolution, max-pooling, and dense layers using ReLU, compared against transfer learning models.	APTOS	95.27%

[19]	2023	Improved base CNN through hyperparameter tuning to create RetNet-10.	APTOS, Messidor2, and IDRiD (combined)	98.65%
[6]		Teacher-student knowledge distillation using ResNet152V2 + Swin Transformer and Xception-CBAM	APTOS/IDRiD	99.04%
[23]		CNN-based staging network predicting multiple DR severity levels	DR Dataset	99.56%
[24]		Ensemble of deep CNN architectures with multi-scale feature extraction	APTOS	79.50%
[20]	2021	Two-stage transfer learning method with PCA and Gradient Boosting for feature reduction and final classification.	E-ophta(training), EyePACS, Messidor.	99.73%

2.3.3 Hybrid Approaches

Hybrid models gained attention as a strong alternative to single CNN by combining different DL models to overcome issues like limited generalization, Lack of Interpretability and class imbalance. Hybrid models include combining networks such as Inception V3 with ResNet50, Dense-Net with Inception blocks, VGG19 with ResNet50 for the extraction of both low-level texture patterns and high-level semantic details which increase model efficiency, and boost diagnostic performance across all DR grades. Hybrid models handle varied image quality and subtle lesion differences more effectively.

Several studies have demonstrated the effectiveness of CNN–CNN feature fusion. A hybrid CNN model [25] extracted deep features by combining ResNet50 and InceptionV3 which were fused prior to classification. Evaluated on the Kaggle DR dataset, the proposed method achieved an accuracy of 96.85%. In another study [26] a hybrid deep learning model VR-FuseNet is proposed by combining two state of the art CNN networks VGG19 and ResNet50V2. To handle limitations like dataset imbalance, diversity and generalization a hybrid dataset combining five publicly available DR datasets (APTOS 2019, DDR, IDRiD, Messidor 2, and Retino) is proposed. Various preprocessing techniques such as Synthetic Minority Over-sampling Technique (SMOTE) and Contrast Limited Adaptive Histogram Equalization (CLAHE) used for class balancing and image enhancement respectably. The authors reported an accuracy

of 91.824%.

Similarly, In [27] a hybrid CNN architecture combining ResNet-16 and GoogleNet based on Adaptive Particle Swarm Optimization (APSO) for feature extraction is proposed. By applying various ML classifiers (random forest, support vector machine, decision tree, and linear regression) for classification the proposed method achieved an accuracy of 94%. In another study [28] a novel hybrid deep learning framework integrating CNNs with Recurrent Neural Networks (RNNs) specifically Long Short Term Memory (LSTM) is proposed for DR early detection and progression respectably. Evaluated on benchmark datasets DRIVE and EyePACS proposed approach achieved an accuracy of 97.5%.

Attention-based hybrid models have further enhanced feature discrimination. The study by [29] which serves as a foundation of this research applied a dual-attention mechanism in deep learning models to enhance DR classification. They utilized three pre-trained networks, namely, MobileNetV3-small, Efficientnet-b0, and DenseNet-169 as the backbone architectures. By combining global attention block (GAB) and category attention block (CAB) the model selectively focuses on clinically relevant regions of fundus images. This dual-attention approach improves classification accuracy and interpretability by highlighting lesion areas through attention heatmaps, which are essential for clinical acceptance. Evaluated on two benchmark datasets, experimental results show that on APTOS dataset, the DenseNet-169 achieved 83.20% accuracy, followed by MobileNetV3-small and EfficientNet-b0, which yielded 82% and 80% accuracies, respectively. On the EYEPACS dataset, the EfficientNet-b0 achieved accuracy of 80%, while the DenseNet-169 and MobileNetV3-small yielded 75.43% and 76.68% accuracies, respectively.

Several studies have combined deep learning models with classical machine learning classifiers to improve efficiency and interpretability. A transfer learning-based hybrid approach combining GoogleNet and ResNet-18 for feature extraction, followed by an SVM classifier, achieved an accuracy of 89.29% on the APTOS dataset [30]. Similarly, In [31] authors presented a novel transfer learning approach using NASNet-Large and ResNet-50. The proposed work applied various image pre-processing and data augmentation techniques to enhance fundus image quality and class imbalance. Tested on the APTOS dataset, the method achieved an accuracy of 91.14%. Another study by [32] presented EffNet-SVM by combining EfficientNetV2 small with Support Vector Machine (SVM) classifier for feature extraction and classification respectably.

Evaluated on APTOS dataset the proposed EffNet-SVM model achieved an accuracy of 97.26% demonstrating that hybrid architectures paired with traditional ML classifiers can yield competitive results.

Likewise the study by [33] proposed a transfer learning-based approach for diabetic retinopathy detection and grading. They utilized pre-trained models (ShuffleNet and ResNet-18), to extract deep features from pooling layers and combined using an Error Correction Output Code (ECOC) ensemble classifier. To enhance performance, they further used Adaptive Differential Evolution (ADE) for simultaneous feature selection and parameter tuning. The proposed model is evaluated on the APTOS and the combined EyePACS + Messidor-2 datasets, achieving 82% accuracy for APTOS 5-class grading, 96% for APTOS 2-class, and 75% for EyePACS + Messidor-2 3-class grading. In [34] authors proposed a hybrid deep learning framework for automatic diabetic retinopathy detection and grading by integrating a CNN with two pre-trained models, VGG16 and VGG19, to learn contextual and visual features. Evaluated on APTOS dataset it achieved 90.60% accuracy. Although hybrid models provide higher classification accuracy but their more computation power, longer training limit their interpretability and efficiency in real time diagnostic environment. The summary of existing hybrid approaches for DR is presented in Table 2.6.

Table 2.6: Hybrid Approaches for Diabetic Retinopathy (2023-2025)

Ref	Year	Method/Approaches	Dataset	Acc (%)
[26]	2026	VR-FuseNet by combining VGG19 + ResNet50V2 / Hybrid Deep Learning approach	Aptos/Messidor-2/DDR/IDRID	91.824%
[28]	2025	CNN+RNN(LSTM)/Feature fusion	DRIVE/EyePACS	97.5%
[29]		CNN with dual attention modules (GAB + CAB) for enhanced feature extraction	APTOS /EyePACS	83.20%
[31]		Transfer learning approach using NASNet-Large and ResNet-50	APTOS	91.14%
[32]		EffNet-SVM model by combining EfficientNetV2 small with SVM classifier	APTOS	97.26%

[27]	2024	Hybrid CNN (ResNet-16 + Google-Net) with APSO + ML Classifiers	EyePACS	94%
[35]		Hybrid deep learning approach combining Self-Attention Network (SAN)+Vision Transformer (Vit)	Kaggle Diabetic Retinopathy Detection dataset	99%
[25]	2023	Combines ResNet50 and InceptionV3 features, fuses them, and classifies DR	Kaggle DR dataset	96.85%
[33]		Pre-trained CNN models + Error Correction Output Code (ECOC) ensemble with Adaptive Differential Evolution (ADE) tuning	APTOS, (EyePac + Messidor-2)	82%
[30]	2022	GoogleNet and ResNet-18 for features extraction with SVM Classifier	APTOS	89.29%
[34]		CNN + two VGG networks (VGG16 + VGG19)	APTOS	90.60%

2.3.4 Vision Transformer Approaches

Recent advancements in DR classification mainly focused on attention mechanisms and Vision Transformers (ViTs) to overcome the limitations of CNN–CNN hybrid models. ViTs can highlight clinically important retinal regions, such as lesions and abnormal blood vessels, instead of processing all areas equally like conventional CNNs vision Transformers analyze images as patches and when combined with CNN feature extractors hybrid models use self-attention to understand global relationships and capture fine local details along with long-range contextual information, leading to better performance under varying image quality and lighting conditions.

A notable contribution in this domain is Hybrid Fusion-Net which integrates a Self-Attention Network (SAN) with a Vision Transformer (ViT) for multi-stage DR detection. Evaluated on the Kaggle DR Detection dataset, the model achieved a high accuracy of 99% [35]. In another study [36] a feature attention module with a pretrained VGG19

convolutional neural network (CNN) is proposed to highlight important retinal regions and fuses them with original features for more accurate DR detection. The proposed attention-based approach achieved 95.70% accuracy on the APTOS dataset, highlighting the benefit of attention mechanisms with CNNs.

Likewise, the study by [37] authors presented unified segmentation classification framework combining U-Net, a pretrained CNN, and a Vision Transformer for lesion segmentation, feature extraction and DR grading respectably. Evaluated on APTOS and IDRiD datasets model achieved overall 89.1% classification accuracy. In another study [38] authors proposed a hybrid deep learning framework combining CNNs for local feature extraction and Transformers for capturing global contextual patterns in retinal fundus images. The proposed method achieved an accuracy of 94.28%. Similarly, in [39] an intelligent DR classification model combining two key modules Feature Extraction Block (FEB) and Grading Prediction Block (GPB) is proposed. Authors used transformer in the FEB and residual attention in the GPB for feature extraction of fundus images and to classify the five stages of DR respectably. Evaluated on two datasets DDR and IDRiD the proposed method achieved an accuracy of 91.54%. The summary of existing vision transformer approaches for DR is presented in Table 2.7.

Table 2.7: Vision Transformer Approaches for Diabetic Retinopathy (2023-2025)

Ref	Year	Method/Approaches	Dataset	Acc (%)
[38]	2025	CNN with Transformers for local and global feature extraction	APTOS	94.28
[35]	2024	hybrid deep learning approach combining Self-Attention Network (SAN)+Vision Transformer (Vit)	Kaggle Diabetic Retinopathy Detection dataset	99%
[37]		Hybrid model combining U-Net segmentation, pretrained CNN, and Vision Transformer (ViT) for lesion segmentation, feature extraction and grading	APTOS/IDRiD	89.1%
[36]	2023	Pretrain VGG19 with feature attention module for highlighting lesion regions	APTOS	95.70%
[39]		Feature Extraction Block (FEB) with transformer and Grading Prediction Block (GPB) with residual attention	DDR/IDRiD	91.54%

Chapter Summary

This chapter presents a comprehensive review of existing research on DR classification, highlighting recent advancements in DL, hybrid CNN and Vision Transformer models. The chosen studies mainly focus on feature extraction, feature selection, attention mechanisms, and classification strategies. Publicly available Benchmark datasets were used mostly, comparative results reported in the literature are also summarized. Based on the limitations of prior work, research gaps are identified. These insights motivate the development of the proposed methodology presented in Chapter 3.

Chapter 3

Proposed Methodology

3.1 Introduction

This chapter presents the proposed methodology consists of different phases such as data preparation, CNN backbone as feature extractor, Multi Stage Attention Block (MSAB), followed by a fully connected and classification layers. This work introduces an efficient and robust framework for automated DR classification using retinal fundus images. The complete framework is shown in Figure 3.1. The detail related to each phase is provided in the subsequent sections.

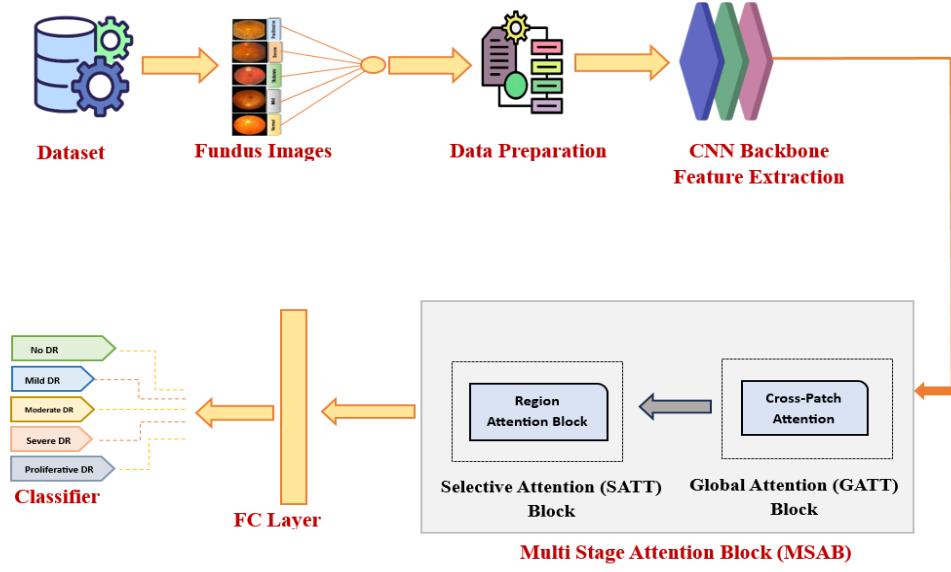


Figure 3.1: Flow Diagram of Proposed Methodology for DR Classification

3.2 Data Preparation

The purpose workflow begins with input of retinal fundus images, which undergo data preparation to ensure consistency in size and quality. These images often suffer from uneven illumination, low contrast, noise, and variability due to different acquisition devices and imaging conditions. Additionally, the available datasets are often imbalanced which leads to binary/multi class imbalance classification problem. To ensure consistent, enhanced and balanced data per class, a comprehensive data preparation phase is required before feature extraction. This phase includes two steps: data augmentation and data preprocessing for synthetic data generation and fundus image enhancement, respectively. These steps ensure that the input data is clean, consistent, and suitable for later phases in the proposed pipeline. Data augmentation is applied to increase dataset diversity and address class imbalance problem. Techniques such as vertical flip, horizontal flip, random affine, brightness adjustment and

crop/zoom are used to create variations of existing images while preserving their medical relevance. This helps the model generalize better, reduces overfitting, and improves performance across different DR stages. Data preprocessing involves tasks such as resizing images to a fixed resolution, normalizing pixel values, and enhancing low quality images to improve clarity. These operations help maintain uniformity across the dataset and allow the model to learn more effectively.

3.2.1 Data Augmentation

In medical image classification tasks such as DR detection, one of the major challenges is the limited diversity of available data along with class imbalance. Certain DR stages, particularly Severe and Proliferative DR, usually have significantly fewer samples compared to more common categories like No DR or Mild DR. If this imbalance is not properly addressed, the model tends to favor the majority classes, resulting in overall poor performance when identifying the minority but clinically critical cases. This can reduce the overall reliability of the system in real-world medical applications. To overcome this limitation, a comprehensive data augmentation step is applied. The objective of this step is to increase the diversity of the dataset by generating synthetic data from existing images while preserving their medical significance. The purpose of oversampling is used to increase the sample size of minority classes. To achieve this, several augmentation techniques such as vertical flip, horizontal flip, random affine and brightness adjustment are applied as shown in figure 3.2 to expand the dataset synthetically. This ensures that the model receives equal number of samples in each class for training and testing.

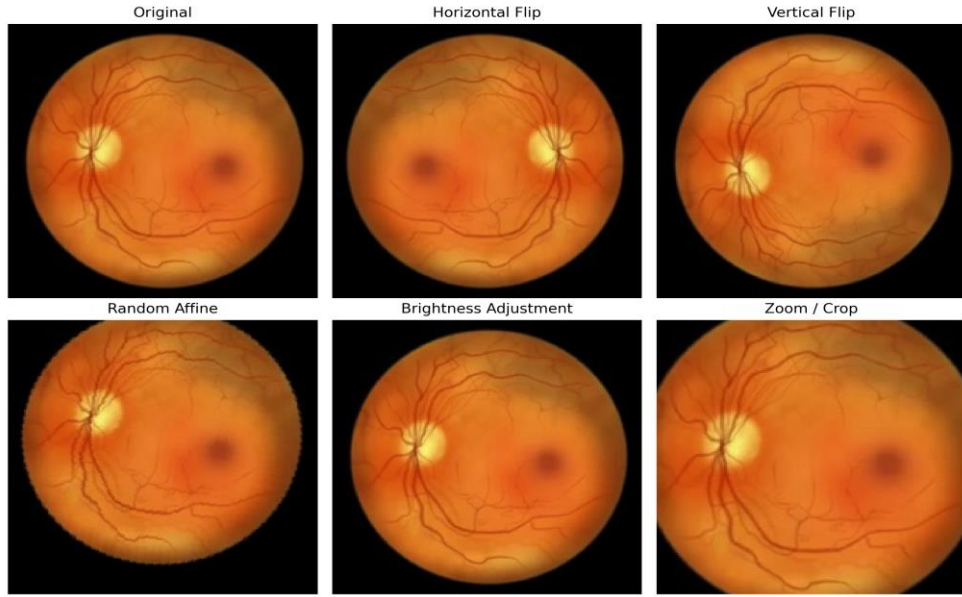


Figure 3.2: Illustration of data augmentation operations applied to retinal fundus images.

3.2.2 Data Preprocessing

Preprocessing of input data is an essential step for image classification and recognition. In this work, all retinal images are preprocessed to ensure consistency and improve visual quality. It also handles illumination changes, light effects, and poor contrast issues in images. In this step, initially each image is resized to a fixed resolution of 224 by 224 pixels with three color channels, allowing uniform input to the CNN architecture EfficientNet-B4. In addition, normalization was applied by scaling pixel values to a standard range, which helps stabilize the training process and improves convergence.

Table 3.1: Description of preprocessing steps for fundus images

Pre-processing Type	Input Image Size	Technique used	Applied To	Output size	Image format	Purpose
Resize	Vary	Resize (224×224)	Train / Val / Test	224×224×3	RGB	Standardize input size
Tensor Conversion		ToTensor()				Convert image to tensor
Normalization		ImageNet Mean & Std				Stable and faster training

3.3 CNN Backbone Feature Extraction

Deep CNNs usually outperform larger-scale databases as compared to small datasets. The pre-trained CNN models are easily accessible to public for use because they offer the learned kernels and weights which can be applied directly or indirectly in different classification [40], segmentation [41], and detection [42] tasks in medical domain. CNN models are mainly trained on large-scale datasets to classify different 1000 object classes. Usually two ways are opted to reuse pre-trained deep CNN models for instance transfer learning with fine-tuning and CNN as deep feature extractor. In this study, the second way CNN as deep feature extractor is applied. Deep feature extraction is a simple procedure in which a single round is required on the input samples to acquire deeply learned information. For this purpose, the EfficientNet-B4 and DenseNet-201 are selected from the pre-trained deep CNN model family.

EfficientNet-B4 is the baseline model in the Efficient-Net family consisting of 237 layers deeply trained on huge amount of data (images) collected from ImageNet database [43]. Model is designed using compound scaling strategy that balances network depth, width, and image resolution to achieve high accuracy with fewer parameters [44]. EfficientNet-B4 (EN-B4) mainly employs Mobile Inverted Bottleneck Convolution (MBConv) blocks with squeeze-and-excitation (SE) modules and skip connections for efficient feature extraction with fewer parameters and to enhance channel-wise feature learning and improve representational power [45]. The model utilizes input size of $224 \times 224 \times 3$ (RGB) and its results provide learned rich information and an estimated value against each class. A schematic view of EfficientNet-B4 architecture is shown in Figure 3.3.

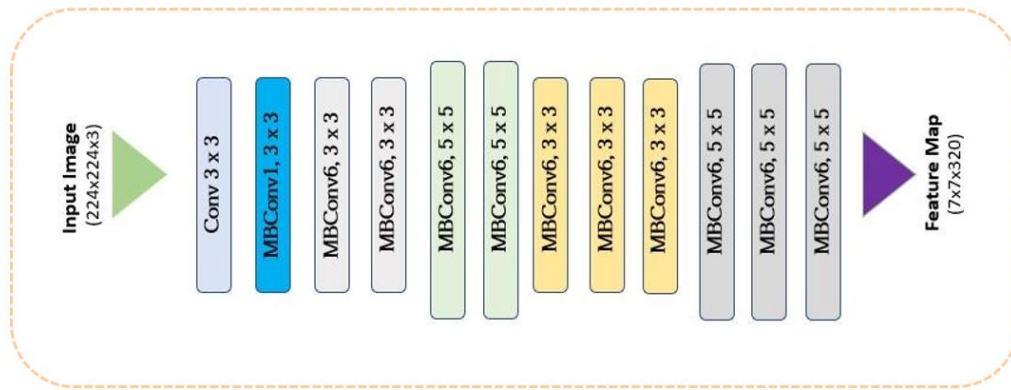


Figure 3.3: Schematic view of EfficientNet-B4 model

DenseNet-201 is used as the baseline model from the Dense Convolutional Network (DenseNet) family, consisting of 201 layers and originally trained on the large-scale ImageNet dataset. Unlike conventional convolutional networks, DenseNet-201 (DN-201) employs dense connectivity, where each layer receives feature maps from all preceding layers through direct connections. Each dense block contains convolutional layers connected via concatenation operations, while transition layers perform feature compression and spatial down-sampling using 1×1 convolutions and average pooling. DenseNet-201 utilizes an input image size of $224 \times 224 \times 3$ (RGB) and progressively extracts hierarchical features, producing rich high-level representations that are finally passed to the classification layer for prediction of diabetic retinopathy classes. A schematic representation of the DenseNet-201 architecture is illustrated in Figure 3.4.

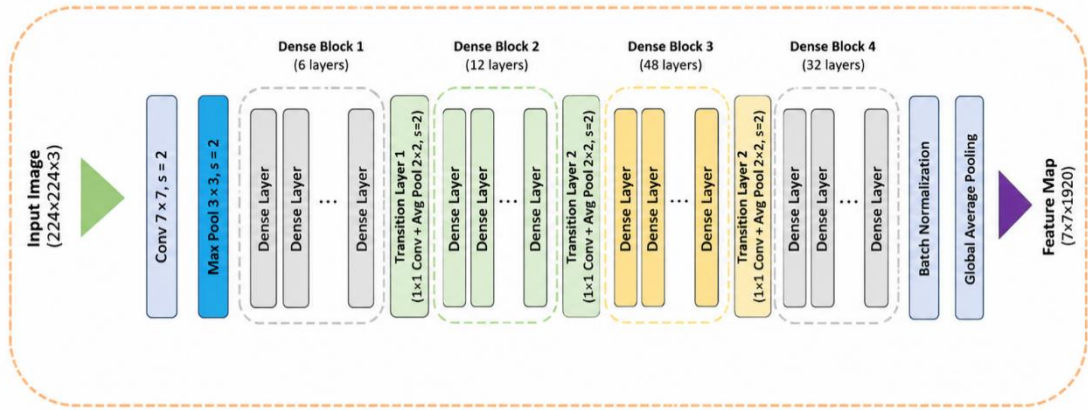


Figure 3.4: Schematic view of DenseNet-201 model

As mentioned above, EF-B4 and DN-201 are selected for deep CNN feature extraction. It is noticeable that the networks with different architectures have strength to capture diverse rich features because of their different depths and structures. Furthermore, an effective ImageNet output does not pass to further assignments, therefore, multiple networks may be required. The CNN backbones consists of multiple convolutional layers interleaved with activation functions and pooling operations. These layers progressively learn low-level features, such as edges and textures, mid-level representations, including vessel patterns, and high-level semantic features corresponding to DR-related lesions. The backbone network serves as a powerful feature encoder, transforming the input image into a rich multi-dimensional feature map. Pre-trained architecture such as EF-B4 and DN-201 are used to leverage transfer

learning, enabling faster convergence and improved generalization, especially in limited-data scenarios.

This allows the model to start with already learned general visual features, reducing training time and improving performance. The original fully connected classification layer is removed so that only the convolutional feature extraction part is used. The input images are resized to a resolution of $224 \times 224 \times 3$ and produces high-dimensional feature maps of $7 \times 7 \times 1792/1920$ that represent different levels of information within the image. These feature maps contain both low-level features, such as edges, textures, and color variations, and high-level semantic information. The overall feature extraction flow is shown in figure 3.5 and 3.6.

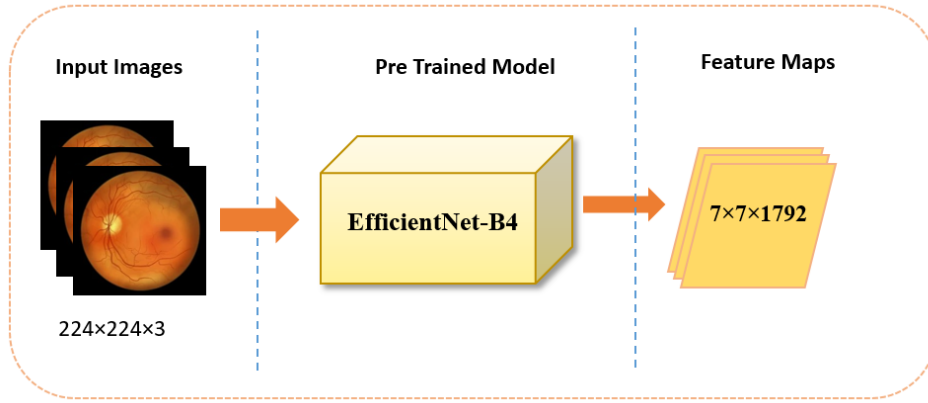


Figure 3.5: Deep CNN feature extraction using pre-trained Efficient-Net B4.

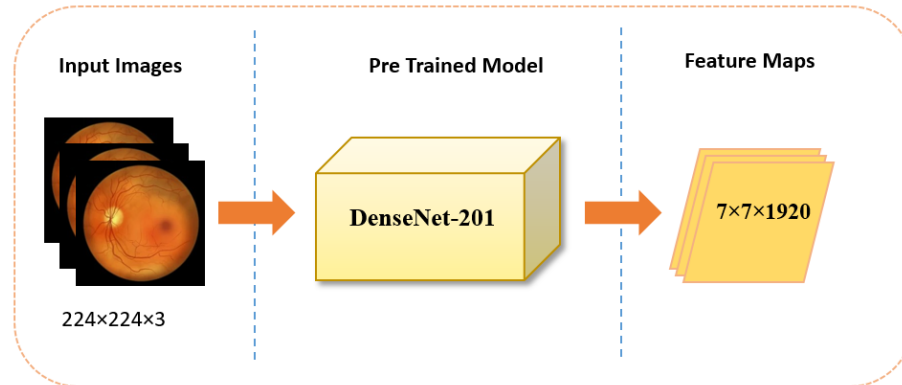


Figure 3.6: Deep CNN feature extraction using pre-trained DenseNet-201.

However, despite their richness, these feature maps may still include redundant or less relevant information. To address this, the extracted features are further refined using the proposed Multi-Stage Attention Module. This module enhances important regions and channels while suppressing noise, ensuring that the model focuses more on clinically relevant features. Additionally, using EF-B4 and DN-201 makes the

framework more practical for real world deployment. Its relatively low computational requirements make it suitable for use in systems with limited resources, such as mobile or edge-based medical applications. This is particularly important for large-scale screening programs where efficiency and speed are crucial. Overall, the combination of features extracted from EF-B4 and DN-201 with the attention mechanism creates a powerful feature extraction pipeline. The backbone provides strong and efficient representations, while the attention module refines these features to improve accuracy and robustness. This integration forms the core of the proposed DR classification system, enabling reliable and effective detection across diverse retinal images.

3.4 Multi-Stage Attention Block (MSAB)

The Multi-Stage Attention Block (MSAB) serves as a key component of the proposed architecture, designed to enhance the discriminative ability of the features extracted by the EF-B4 and DN-201 backbones. Backbone produces high-dimensional feature maps that include useful clinical information such as microaneurysms, hemorrhages, and exudates with less background patterns, which is often difficult to detect in DR classification. Despite their ability to capture rich useful information, conventional CNNs treat all features uniformly; which may fail to give sufficient importance to clinically significant regions.

To address this issue, the proposed Multi stage attention block refines the extracted features by focusing more on informative regions and suppressing less useful information. The MSAB is designed in two sequential stages, Global Attention Block (GATT) and Selective Attention Block (SATT) to capture the similarity between patches and then focusing on more clinically important regions required for accurate classification of DR stages together forming a dual-stage attention mechanism. This design allows the model to capture both fine-grained details and global retinal context to distinguish between closely related DR stages. In addition to improving accuracy, MSAB is designed computationally efficient by operating directly on the feature maps produced by the backbones EF-B4 and DN-201 without requiring heavy architectural modifications. This makes the model lightweight and suitable for real-world deployments, where both performance and efficiency are equally important. Overall, the integration of the MSAB with EF-B4 and DN-201 backbones creates a balanced and effective framework making it well-suited for practical DR detection systems.

3.4.1 Global Attention (GATT) Block

The Cross-Patch Attention block is designed to capture the similarity between different regions of feature map produced by backbones. In DR, abnormalities are often scattered across multiple areas of the retina which may indicate a more advanced stage of the disease. Since standard CNNs often focus only on local patterns due to their small receptive fields it is important to analyze the image as a whole and capture relationships between similar regions. Without understanding these global relationships, the model may overlook important contextual information that is essential for accurate diagnosis. To address this limitation, Cross-Patch Attention block is introduced to capture global interaction between different patches of the feature map. The overall workflow of GATT block is shown in Figure 3.7.

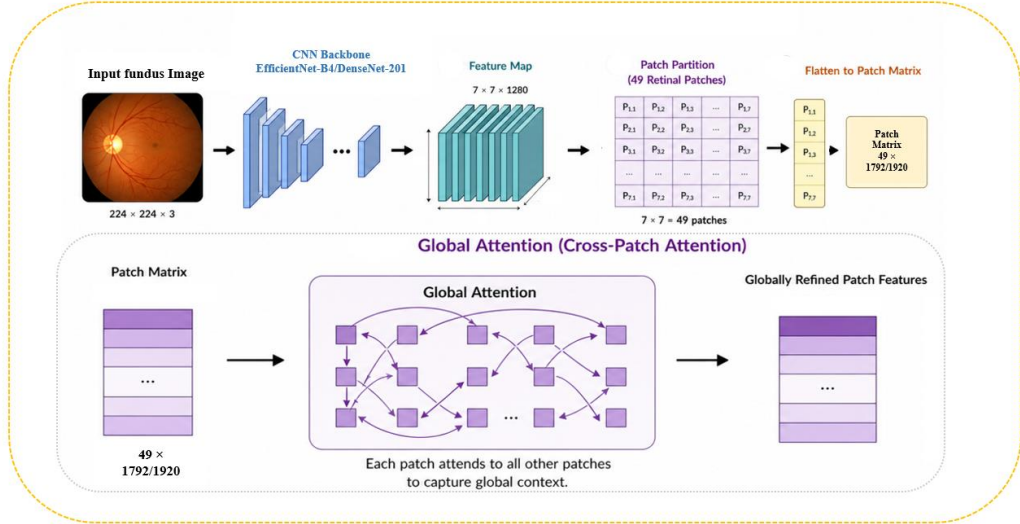


Figure 3.7: Global Attention (GATT) block illustrating global contextual relationships using cross-patch attention.

The workflow of Global Attention Block begins with input image size of $224 \times 224 \times 3$, which is first passed through the pre trained CNN backbone, namely EF-B4 and DN-201 to extract deep feature representations. The backbones produced a high-level feature map $7 \times 7 \times 1792/1920$ containing spatial and semantic information related to clinically important regions. This feature map serves as the input to the GATT block. Then, the extracted feature map is partitioned into 49 spatial regions referred to as patches. Each patch represents 1 spatial location by a $1792/1920$ dimensional feature vector, allowing the model to analyze relationships between spatial regions. After partitioning, the patches are rearranged and flattened to form a patch matrix, where each

row corresponds to a patch feature vector. The patch matrix is then fed into the proposed GATT block where each patch interacts with all other patches to learn inter-region relationships. Instead of treating each region independently, the attention mechanism computes similarity between patches and assigns adaptive importance based on their contextual relevance. Due to this global interaction between distant regions, the model improves prediction stability and generalization under various illumination conditions, image quality differences and noise commonly present in real-world medical datasets. The GATT block focuses solely on spatial context learning and produces globally refined feature representations that are subsequently forwarded to the next attention stage SATT for further feature refinement.

3.4.2 Selective Attention (SATT) Block

The Regional Attention Block is designed to improve the quality of channel wise feature representations by focusing more on features extracted from GATT block. Unlike GATT block which focuses on learning spatial relationships between different regions, SATT works at the channel level, to capture the most useful features for DR classification. Although GATT block captures the global relationships between regions and contain clinically rich features but still contain many irrelevant background patterns that leads to poor classification. To address this problem SATT block is introduced to focus only on clinically relevant feature channels for accurate analysis. The overall workflow of SATT block is shown in figure 3.8.

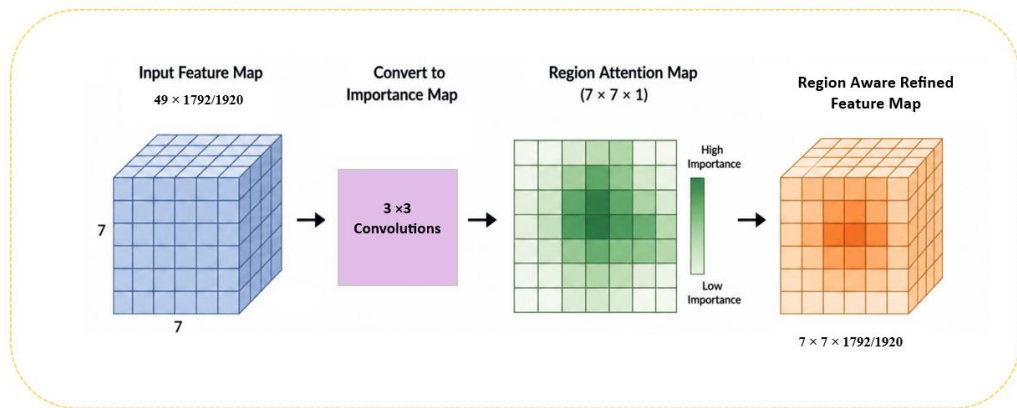


Figure 3.8: Selective Attention (SATT) block highlighting channel-wise feature refinement through channel-wise attention weighting.

The Selective Attention (SATT) block receives the input of globally refined feature map produced by GATT block. This feature map has a spatial dimension of 7×7 with 1792/1920 feature channels. Although spatial relationships have already been refined in the previous attention stage, but still all feature channels contribute uniformly, which may lead to poor classification problem because it contains both relevant and less relevant patterns. Therefore, channel-wise refinement is performed to emphasize meaningful feature representations.

First, the refined feature map from the GATT block is passed to a 3×3 convolution operation that produces an intermediate attention map of size $7 \times 7 \times 1$, which summarizes the aggregated contribution of channel-wise features at each spatial position. Then the produced attention map is normalized to form the selective attention map. This attention map assigns high weights to the most important features selected by GATT block and lower values to less useful background patterns. Finally, the output of the SATT block is a region-aware feature map with the dimensional size of $7 \times 7 \times 1792/1920$ which contains enhanced channel-wise features and are subsequently forwarded to the classification layers. Overall, the sequential combination of GATT and SATT blocks in MSAB enhances the spatial and channel information for final DR classification.

3.5 Fully Connected Layer

The final stage of the proposed framework performs DR classification using refined feature map of size $7 \times 7 \times 1792/1920$ obtained from MSAB. The extracted rich features are first flattened into one-dimensional feature vector and then passed to a Fully Connected (FC) layer, where these high-dimensional features are transformed into classification scores. These scores reflect the model's confidence for each class. The output layer contains five neurons representing the DR severity levels, No DR, Mild DR, Moderate DR, Severe DR, and Proliferative DR. A Softmax activation function is applied at the final layer to transform the scores into probability values between 0 and 1, where each value indicates that the input image belongs to a particular class. The overall workflow from MSAB feature extraction to final classification is illustrated in figure 3.9.

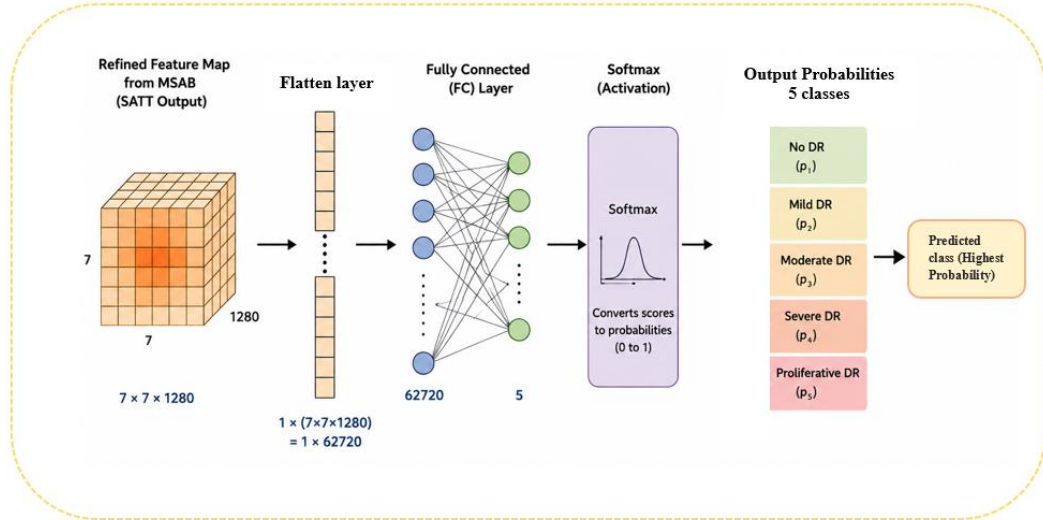


Figure 3.9: Overall workflow illustrating final diabetic retinopathy prediction through flattening, fully connected layers, and Softmax classification.

3.6 Classification Layer

The classification layer is the final part of the proposed framework, where the output scores obtained from the FC layer are converted into class probabilities using the Softmax activation function. Softmax normalizes the probabilities between 0 and 1, for final prediction. These probabilities show how confident the model is about each DR stage. Then the class with the highest probability is chosen as the final prediction. During training, these predicted probabilities are compared with the ground truth labels, which helps the model learn and gradually improve its performance towards its actual class. Overall, the classifier helps turn the learned features into meaningful DR stage predictions to improve accuracy.

Chapter 4

Results and Discussions

4.1 Introduction

This chapter presents the experimental setup and results obtained after implementing the proposed methodology for DR classification (as discussed in Chapter 3). It includes the dataset description, evaluation metrics, experimental settings, confusion matrix analysis, ablation study, and comparison with existing methods. The results obtained from different experiments are analyzed in detail to validate the effectiveness of the proposed approach. The details related to each section is provided in the subsequent sections.

4.2 Datasets Description

Datasets are required to train, validate, and to evaluate the performance of the proposed framework using labeled retinal fundus images representing different stages of DR, enabling the model to learn disease-specific areas [46]. The proposed framework is evaluated using two publicly available benchmark retinal image datasets, APTOS and IDRiD, widely used for DR severity classification. The details about these datasets is provided in subsequent sections.

4.2.1 APTOS Dataset

The Asia Pacific Tele-Ophthalmology Society (APTOS) dataset was released as part of the Kaggle DR Detection Challenge in 2019 [47]. Overall, the APTOS dataset consists of 3,669 images. The primary objective of using the fundus imaging dataset was to assess the severity of the disease to one of five clinically defined DR categories. These categories include: No DR, Mild DR, Moderate DR, Severe DR, and Proliferative DR. The APTOS dataset serves as a compact yet diverse benchmark for evaluating the classification capability of deep learning models.

4.2.2 IDRid Dataset

The Indian diabetic retinopathy image (IDRiD) dataset [48] provides 1548 high-quality images with clearer lesion visibility and more consistent imaging conditions, making it suitable for controlled evaluation. By using both datasets, the model is exposed to a wider variety of images. This helps it learn different patterns more effectively. As a result, the model can handle variations better and perform well on new unseen data. The overall statistics of both datasets is shown in table 4.1.

Table 4.1: Statistics of Applied Datasets

Datasets	Image type	No of images	Challenges
APTOS	Retinal Fundus Images	3,662 images	Small dataset size, Imbalance classes, Lesion Similarity
IDRiD	Retinal Fundus Images	1548 images	Class imbalance, low resolution, limited samples, uneven distribution

4.2.3 Data Oversampling

Oversampling is a data balancing technique used to address unequal class distribution in machine learning datasets [49]. It works by increasing the number of samples in minority classes so that all classes contribute equally during model training. Instead of collecting new data, oversampling generates additional samples through duplication of original images. In medical image analysis, particularly in DR grading, dataset imbalance is a common challenge. Certain disease stages such as Mild, Severe, or Proliferative DR typically contain significantly fewer samples compared to the No DR class as shown in table 4.2. This class imbalance causes model biasness towards majority classes, resulting in poor sensitivity for clinically important minority stages. To overcome this issue, an oversampling strategy is applied to both APTOS and IDRiD datasets. Initially, class-wise image distributions were analyzed to identify minority categories. Result shows that in the APTOS dataset, all five classes DR(0), DR(1), DR(2), DR(3) and DR(4) contains 1805, 370, 999, 193, 295 sample images, respectively. Similarly, the IDRiD datasets contain 402, 60, 408, 222 and 147 images belonging to each class.

Table 4.2: Oversampling statistics for (APTOS, IDRiD) datasets

Description	Datasets	Classes				
		No DR	Mild	Moderate	Severe	Proliferative
Class-wise selected images	APTOS	1805	370	999	193	295
Oversampled images		0	1435	806	1612	1510
Original + Oversampled images		1805 in each class				
Class-wise selected images	IDRiD	402	60	408	222	147
Oversampled images		6	348	0	186	261
Original + Oversampled images		408 in each class				

Additional samples were generated for underrepresented classes until all classes reached a uniform sample size. For the APTOS dataset, each DR grade was balanced to contain 1805 images per class after oversampling. Similarly, the IDRiD dataset was balanced to 408 images in each class as shown in figure 4.1 and 4.2. This balancing process ensures that the model receives equal representation from all DR stages during training.

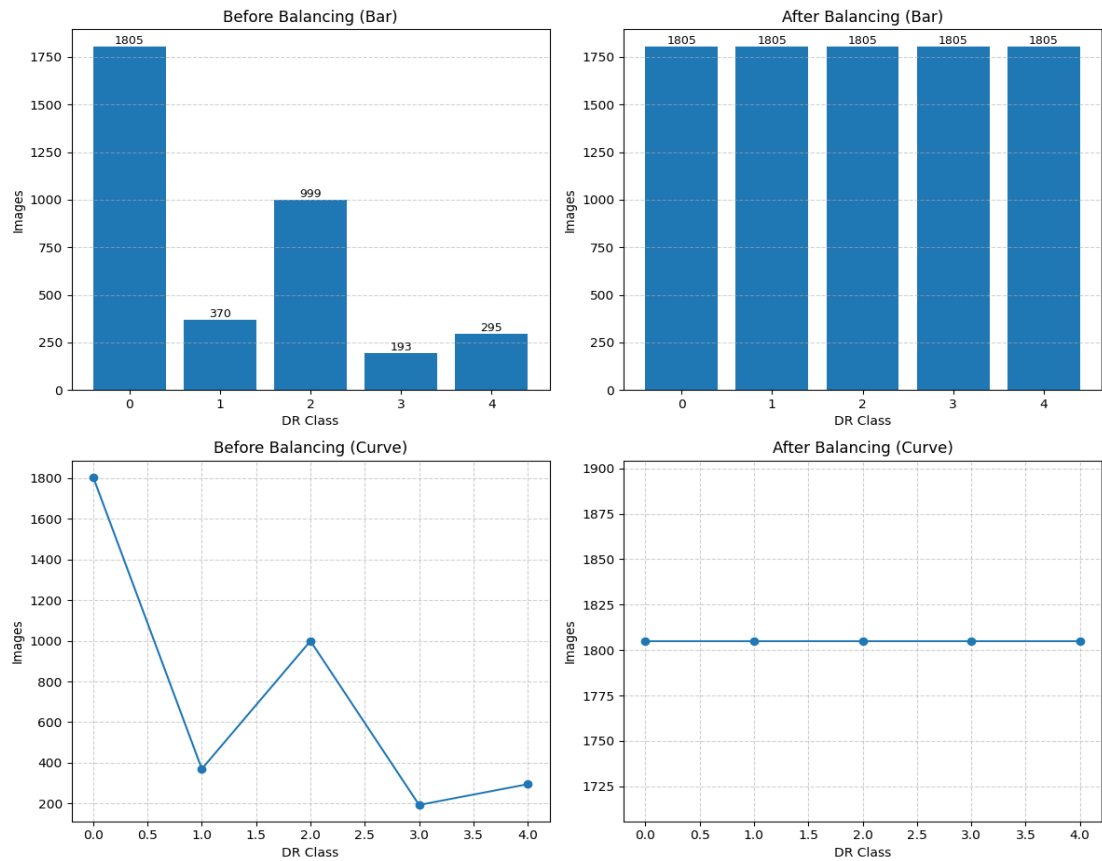


Figure 4.1: APTOS dataset before and after oversampling

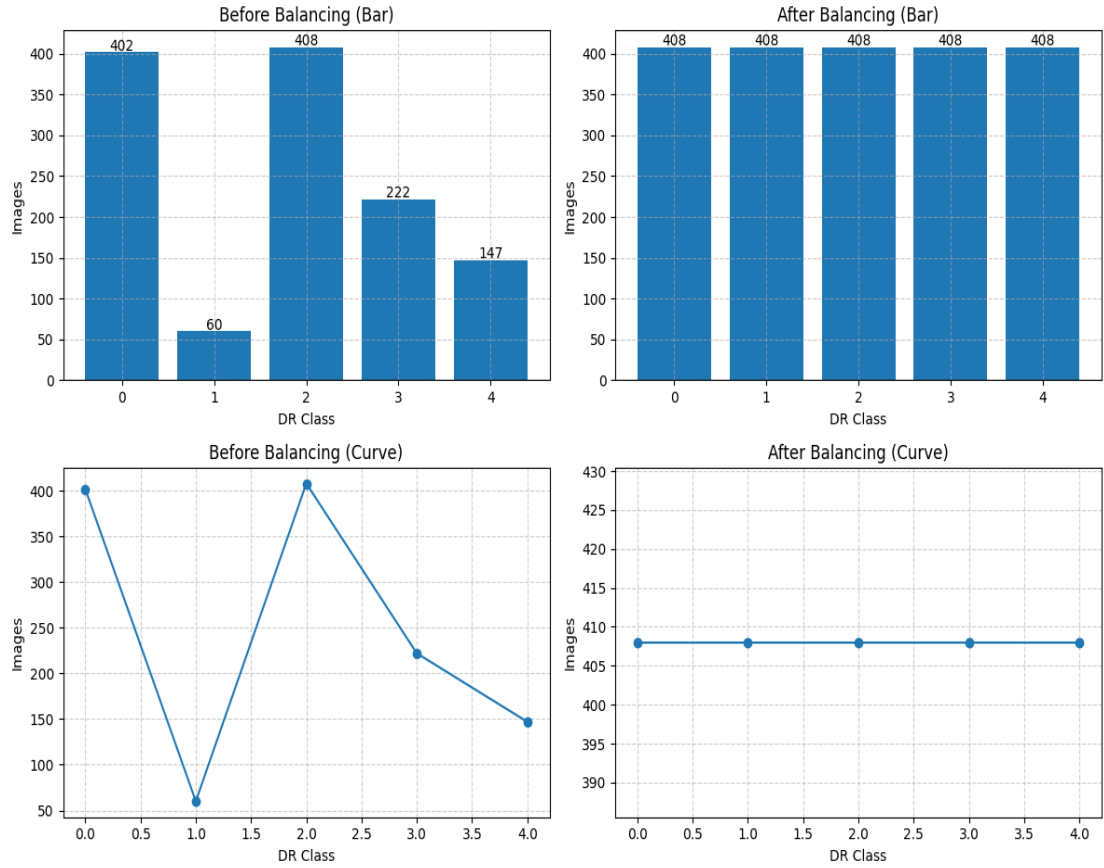


Figure 4.2: IDRiD dataset before and after oversampling

4.3 Evaluation Metrics

To evaluate the performance of the proposed architecture, several evaluation metrics are used to provide a comprehensive analysis under different experimental conditions. First, accuracy is used to measure the overall correctness of the model by calculating correctly predicted values. However, since accuracy alone may be misleading in imbalanced datasets precision and recall are also considered. Precision (specificity) shows how many of the predicted positive DR cases are actually correct, while recall (sensitivity) indicates how well the model can identify all relevant DR cases. Since both metrics are important, the F1-score is used to combine them into a single balanced metric, making it more reliable for imbalanced data. Furthermore, Quadratic Weighted Kappa (QWK) is used as a key metric because it considers the ordered nature of disease stages and penalizes larger misclassifications more heavily. Overall, evaluation metrics work together to provide a reliable and detailed evaluation across both balanced and

imbalanced datasets. The evaluation matrices formulas and experimental parameter setting is shown in table 4.2 and 4.3.

Table 4.3: Experimental Matrices Measure Equations

Matrics	Equations
Accuracy	$\frac{(TP + TN)}{(TP + FP + TN + FN)}$
Precision	$\frac{(TP)}{(TP + FN)}$
Recall	$\frac{(TN)}{(TN + FP)}$
F1-Score	$\frac{TP}{TP + \frac{1}{2}(FN + FP)}$
Quadratic Weighted Kappa (QWK)	$1 - \frac{\sum_{i,j} w_{ij} O_{ij}}{\sum_{i,j} w_{ij} E_{ij}}$

The TP/TN indicates True Positives/Negatives and FP/FN indicates False Positives/Negatives, respectively. Where, W(i,j) and E(i,j) is the pre-defined weight for agreement between raters i and j, O(i,j) is the observed agreement, is the expected agreement by chance. The resulting K value ranges from $[-1, +1]$, with +1 indicating excellent agreement, 0 indicating chance agreement, and negative values indicate better than random chance agreement.

Table 4.4: Experimental Parameter Settings

Metric	APTOS (Imbalanced)	APTOS (Balanced)	IDRiD (Imbalanced)	IDRiD (Balanced)
Accuracy	0.4355	0.6975	0.8093	0.8824
Precision	0.5632	0.7673	0.8136	0.8786
Recall	0.4355	0.6975	0.8093	0.8824
F1-Score	0.4472	0.7095	0.8010	0.8790
QWK	0.7439	0.8013	0.8849	0.9457

4.4 Experimental Settings

Experiments were conducted using a GPU-enabled Google Colab environment. The proposed model integrates EfficientNet-B4 as the backbone with MSAB. EfficientNet-B4 was selected as the backbone network due to its deep feature learning capability and computational efficiency. The datasets were divided into training, validation, and testing subsets to ensure balanced representation of all DR severity classes using a stratified split ratio of 80%, 10%, and 10%, respectively. All retinal images were resized to 224×224 pixels to maintain consistent input dimensions to backbone. Data augmentation techniques, including horizontal flipping, vertical flipping and random affine were applied to increase data diversity. To address class imbalance, Random Over-Sampling was applied to the training data. The model was trained using the AdamW optimizer, learning rate 1×10^{-4} , batch size of 16 with weighted cross-entropy loss function to assign greater importance to minority classes. The best-performing model was selected based on validation performance using the Quadratic Weighted Kappa (QWK) score, which is particularly suitable for ordinal classification tasks such as DR grading. The integration of EfficientNet-B4 with the MSAB enhanced feature representation and improved discrimination between different DR severity levels. Overall, the proposed experimental setup provided a stable and effective training resulting in reliable performance across both balanced and imbalanced datasets.

Table 4.5: Experimental Parameters Setting

Parameter	Setting	Parameter	Setting
Development Environment	Google Colab (GPU-enabled)	Optimizer	AdamW
Backbone Network	EfficientNet-B4	Learning Rate	1×10^{-4}
Input Image Size	224×224	Batch Size	16
Datasets Used	APTOS and IDRiD	Model Selection Metric	Quadratic Weighted Kappa (QWK)
Train/Validation/Test Split	80% / 10% / 10%	Classification Task	Five-Class DR Classification
Data Augmentation	Horizontal Flip, Rotation	Class Balancing Method	Random Over-Sampling

4.4.1 Train/Val/Test data balancing

The class distribution of retinal images before and after balancing is illustrated in Figures 4.3–4.6. Initially, both the APTOS and IDRiD datasets showed significant class imbalance, where majority classes like No DR and mild contained substantially more samples than minority classes. Such imbalance can bias model learning towards dominant classes. To overcome this issue, Random Over-Sampling is applied during training to increase the representation of minority DR stages and to improve the overall class distribution. As a result, the model learned more balanced features across all DR stages and achieved improved classification performance, particularly for severe DR stages. Figures 4.3 and 4.5 present the original class distributions of APTOS and IDRiD datasets before balancing, while Figures 4.4 and 4.6 show the balanced distributions obtained after applying the oversampling technique.

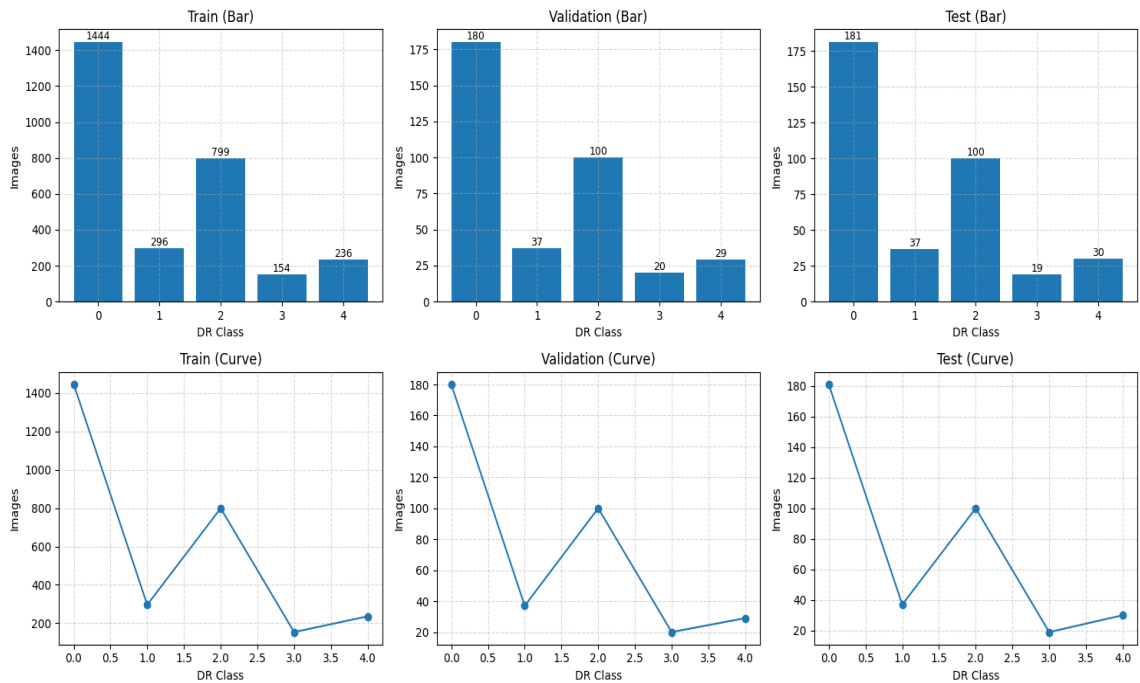


Figure 4.3: APTOS Train\Val\Test data before balance

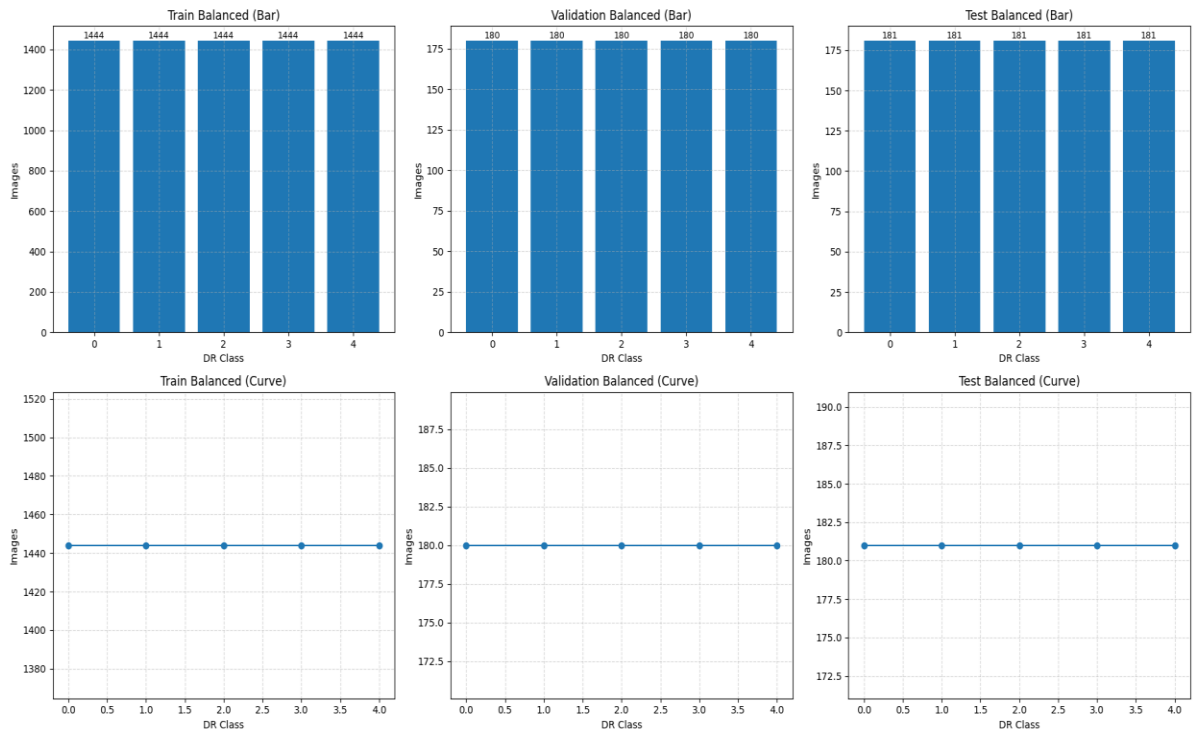


Figure 4.4: APTOS Train\Val\Test data after balance

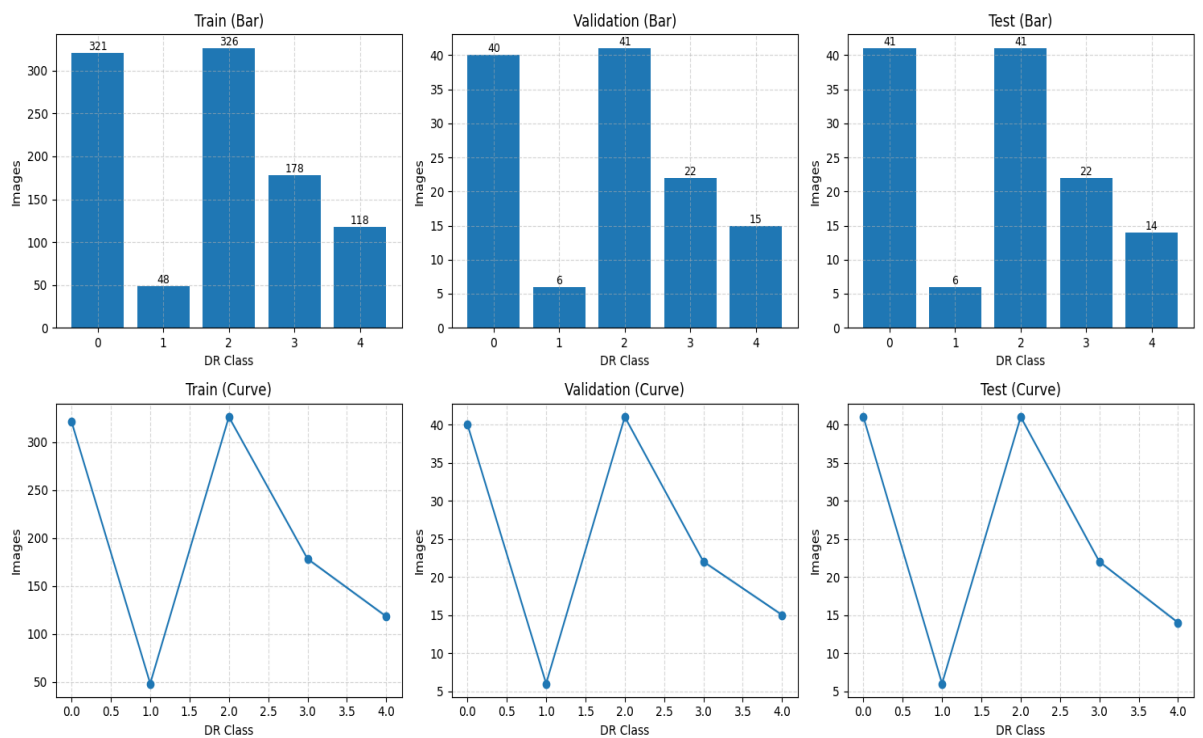


Figure 4.5: IDRiD Train\Val\Test data before balance

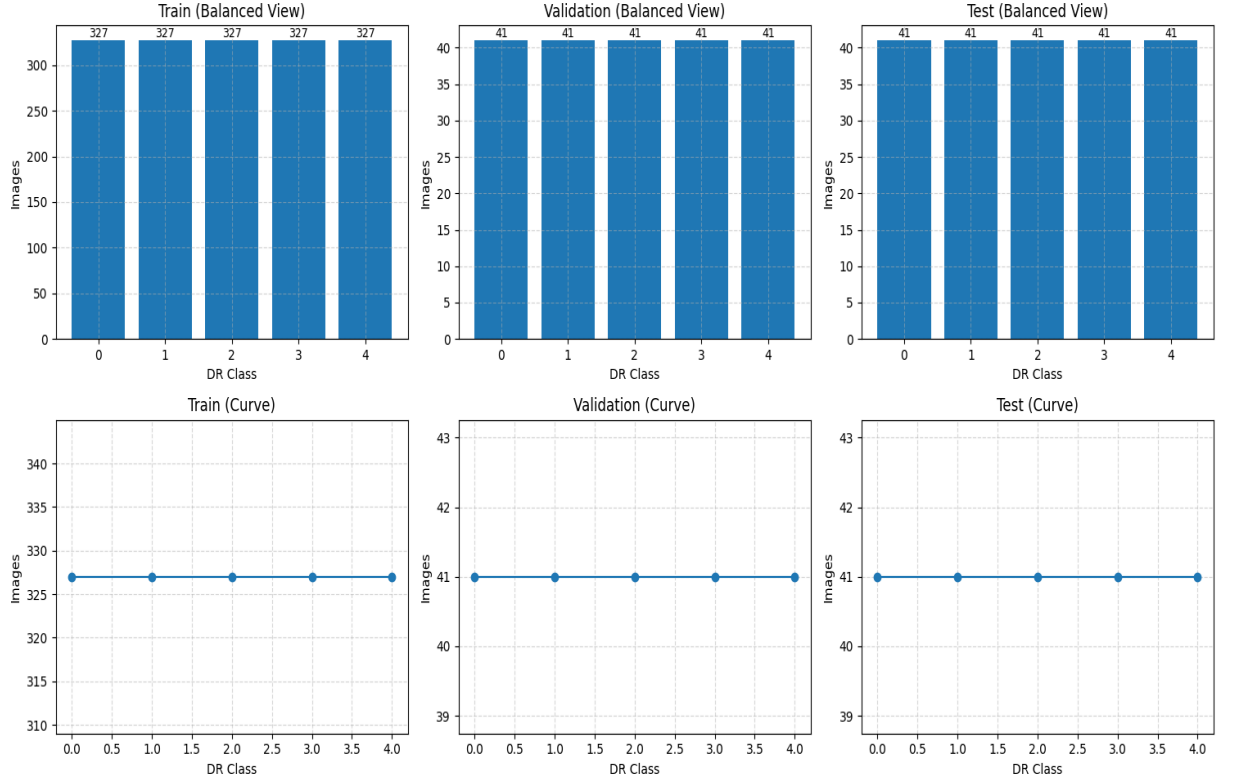


Figure 4.6: IDRIID Train\Val\Test data after balance

4.5 Confusion Matrix Analysis

To further evaluate the performance of the proposed model, confusion matrices are used to analyze predictions for each DR class individually. The confusion matrix provides a detailed representation of both correct and incorrect classifications across different disease stages. Figure 4.7– 4.10 present the confusion matrices obtained from the APTOS and IDRIID datasets with EfficientNet-B4 backbone under both imbalanced and balanced settings. Figure 4.11– 4.14 present the confusion matrices obtained from the APTOS and IDRIID datasets with DenseNet-201 backbone under both imbalanced and balanced settings. The diagonal elements of each matrix represent correctly classified samples, while the off-diagonal elements indicate misclassification. The results show that balancing the datasets improves classification performance by reducing confusion among minority classes. In particular, the balanced datasets demonstrate stronger diagonal dominance, indicating more accurate predictions across all DR severity levels.

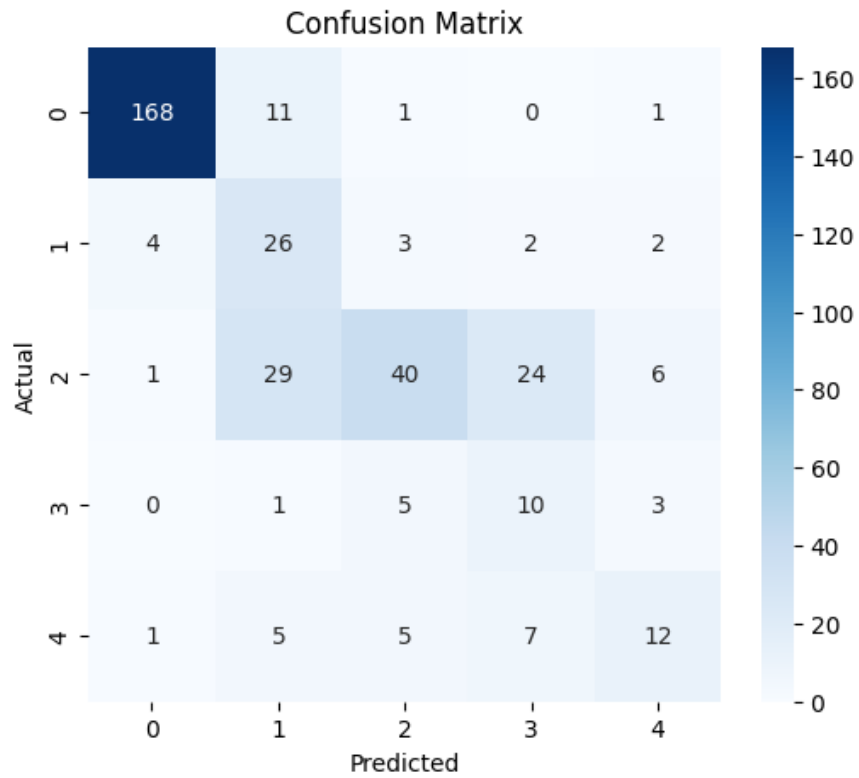


Figure 4.7: Confusion Matrix for APTOS Imbalanced using EfficientNet-B4

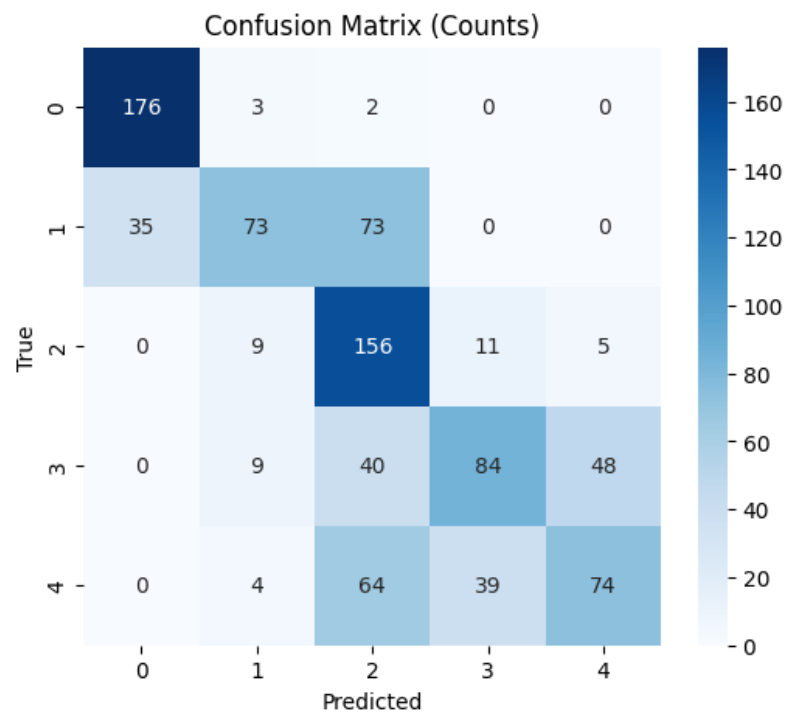


Figure 4.8: Confusion Matrix for APTOS Balanced using EfficientNet-B4

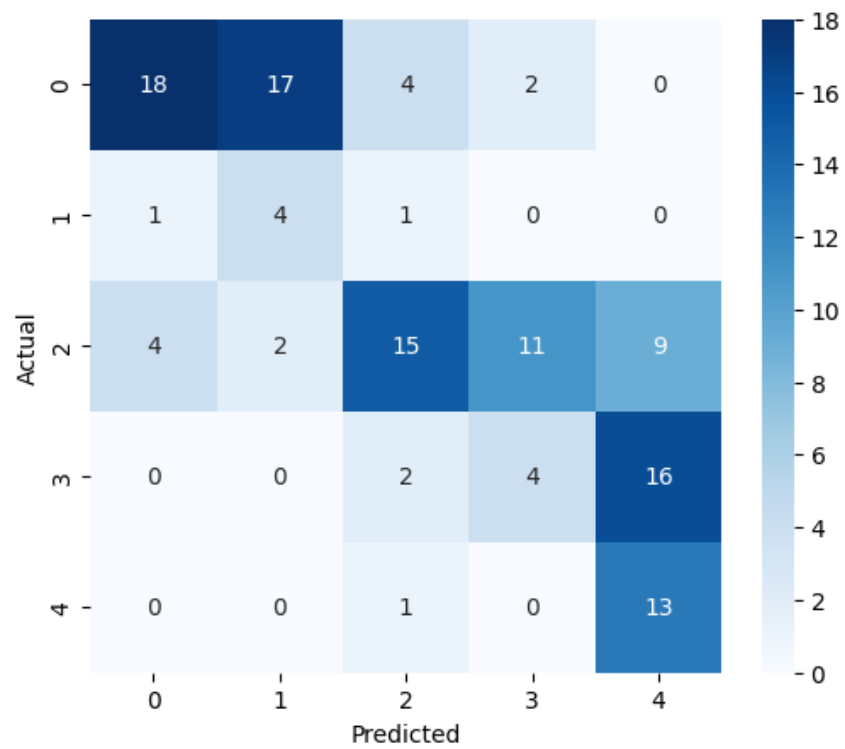


Figure 4.9: Confusion Matrix for IDRiD Imbalanced using EfficientNet-B4

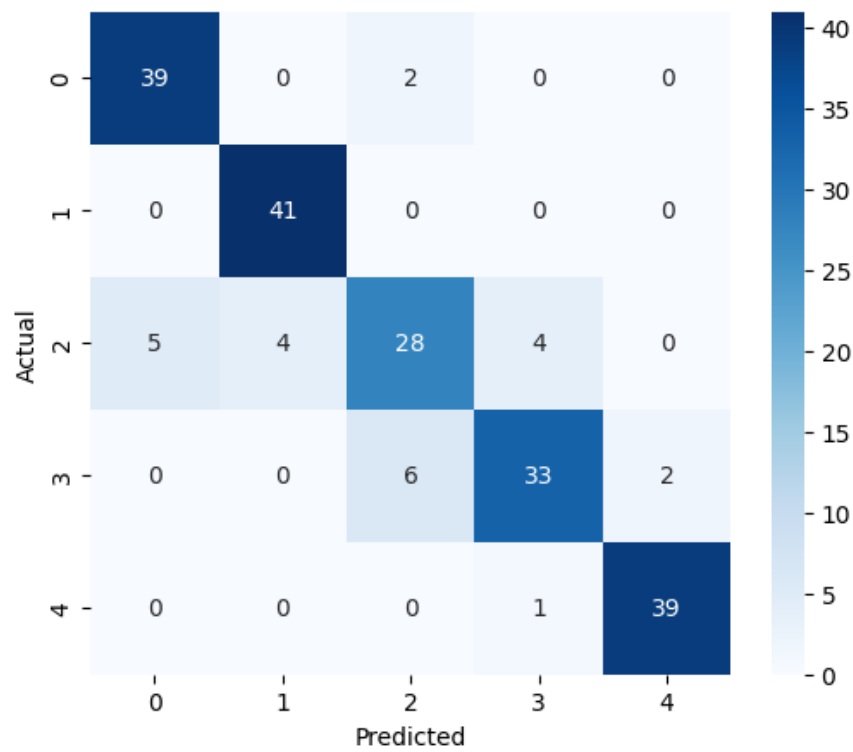


Figure 4.10: Confusion Matrix for IDRiD Balanced using EfficientNet-B4

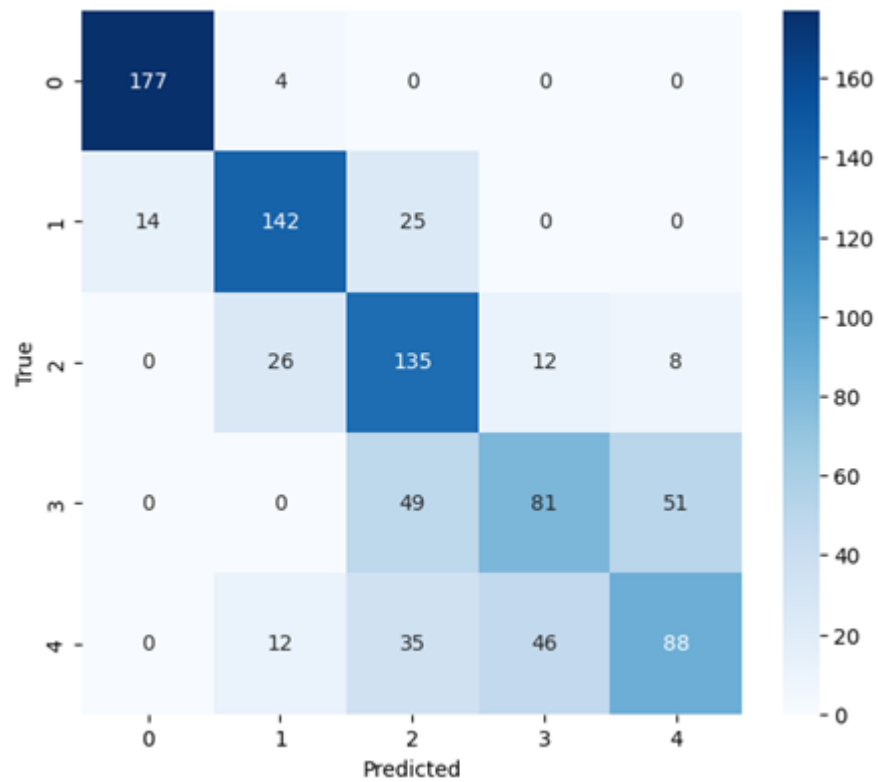


Figure 4.11: Confusion Matrix for APTOS Imbalanced using DenseNet-121

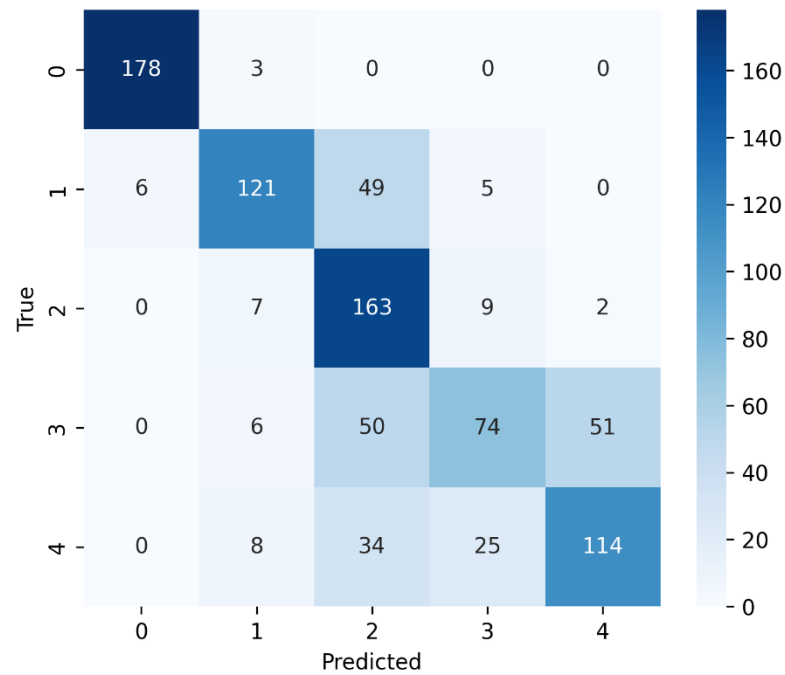


Figure 4.12: Confusion Matrix for APTOS balanced using DenseNet-201

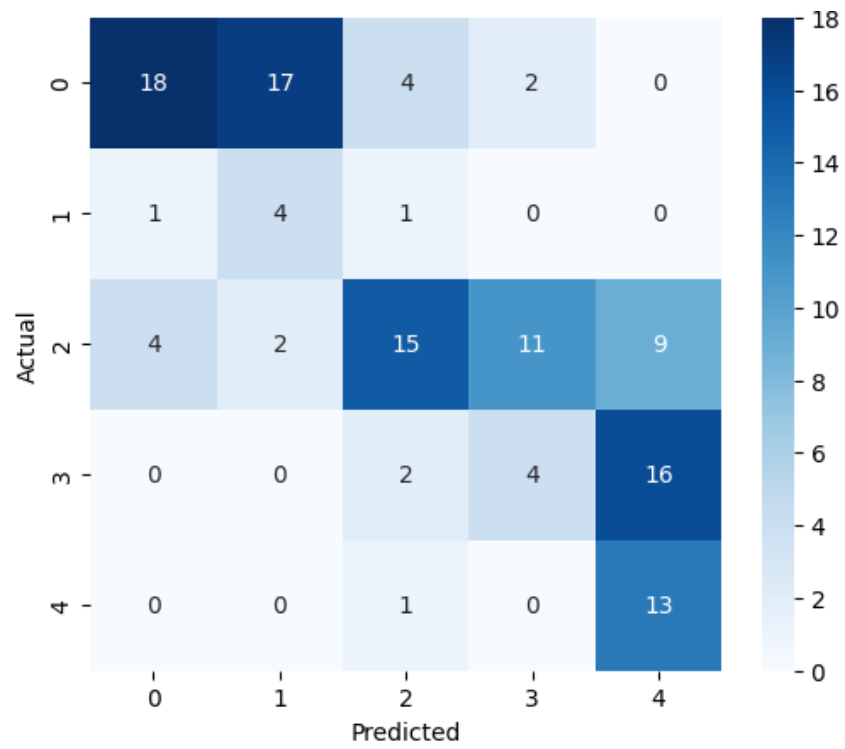


Figure 4.13: Confusion Matrix for IDRiD ImBalanced using DenseNet-201

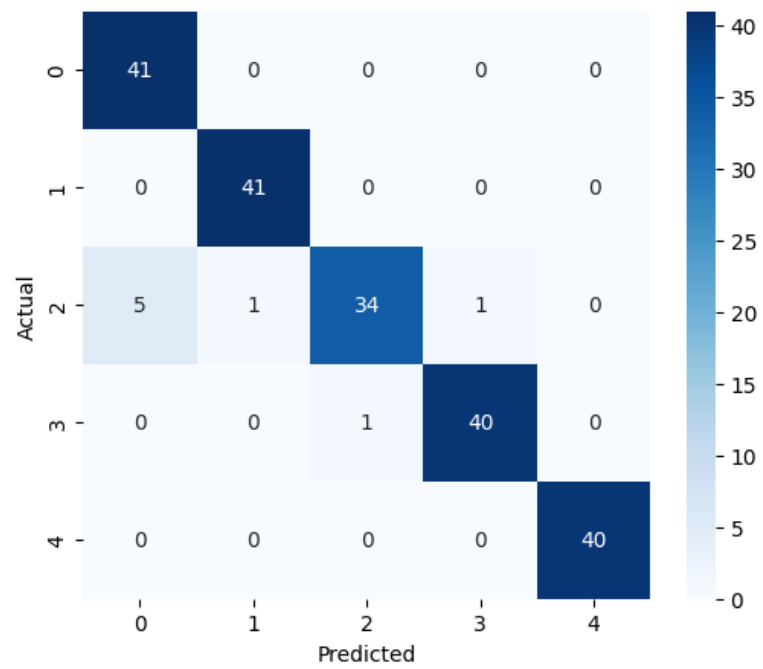


Figure 4.14: Confusion Matrix for IDRiD Balanced using DenseNet-201

4.6 Ablation Study

To evaluate the effectiveness of the proposed architecture, an ablation study is performed. Initially, EfficientNet-B4 and DenseNet-201 are used as the baseline models. Subsequently, experiments were performed using individual Global Attention (GATT) and Selective Attention (SATT) modules with the backbones, followed by evaluation of the complete proposed architecture with both balanced and imbalanced datasets. This analysis helps in understanding how each module affects overall classification performance and contributes to feature learning. It also provides insight into the effectiveness of integrating different attention mechanisms within the CNN-based architecture for DR classification. The details related to each section is described in subsequent sections.

4.6.1 Results with Backbone

The experimental results obtained using only the EfficientNet-B4 backbone model are presented in Table 4.6 for both balanced and imbalanced APTOS and IDRiD datasets. The backbone model achieved better performance on balanced datasets as compared to imbalanced, indicating the impact of class distribution on model learning. The balanced APTOS dataset produced higher Accuracy, F1-score, and QWK values than the imbalanced case. Similarly, performance on the IDRiD dataset significantly decreased under the imbalanced setting, highlighting the limitations of using only the backbone network without additional attention mechanisms. The experimental results obtained using only the DenseNet-201 backbone are presented in Table 4.7 for both balanced and imbalanced APTOS and IDRiD datasets. For the APTOS dataset, the balanced setting achieved higher Accuracy, F1-score, and QWK values compared to the imbalanced configuration. This indicates that DenseNet-201 effectively learns discriminative retinal features when sufficient samples from all classes are available. However, performance decreases in the imbalanced scenario due to biased learning toward dominant classes. Similarly, on the IDRiD dataset, the DenseNet-201 backbone achieved excellent results in the balanced setting, reflecting its strong feature reuse and deep representation learning capability. In contrast, a significant reduction in performance is observed for the imbalanced dataset, highlighting the limitations of relying solely on the backbone architecture without incorporating attention mechanisms or imbalance-handling strategies.

Table 4.6: Experimental results of balanced and imbalanced APTOS and IDRiD datasets with EfficientNet-B4 backbone using Accuracy, F1-score, and QWK evaluation metrics.

Dataset	Method	Accuracy	F1 score	QWK
APTOS(Balanced)	Backbone	0.8090	0.8065	0.8441
APTOS(Imbalanced)		0.7432	0.7409	0.8159
IDRiD (Balanced)		0.7661	0.7686	0.8774
IDRiD (Imbalanced)		0.5215	0.5141	0.6407

Table 4.7: Experimental results of balanced and imbalanced APTOS and IDRiD datasets with DenseNet-201 backbone using Accuracy, F1-score, and QWK evaluation metrics.

Dataset	Method	Accuracy	F1 score	QWK
APTOS(Balanced)	Backbone	0.7705	0.7770	0.8317
APTOS(Imbalanced)		0.7156	0.7634	0.7856
IDRiD (Balanced)		0.9435	0.9434	0.9788
IDRiD (Imbalanced)		0.5234	0.6134	0.6735

4.6.2 Results with Cross Attention and Backbone

The experimental results obtained using Cross Attention (CA) with EfficientNet B4 backbone are presented in Table 4.8. The integration of the CA module improves the model ability to focus on discriminative retinal regions, leading to enhanced classification performance. For the APTOS dataset, accuracy and F1-score remains relatively similar to the backbone baseline. Although the QWK score shows a slight improvement in the balanced setting, Therefore, the addition of Cross Attention provides only limited performance gain on this dataset. In contrast, CA with backbone outperformed on IDRiD balanced dataset in terms of Accuracy, F1-score, and QWK as compared to the backbone-only results. The experimental results obtained using Cross Attention (CA) integrated with the DenseNet-201 backbone are presented in Table 4.9 for both balanced and imbalanced APTOS and IDRiD datasets. For the APTOS dataset, the balanced setting achieved better Accuracy, F1-score, and QWK compared to the imbalanced configuration, demonstrating that class balance positively influences

learning performance. Although performance decreases under class imbalance, the CA + backbone model still maintains stable classification capability. On the IDRiD dataset, strong performance is achieved in the balanced scenario, indicating effective feature refinement through cross-attention learning. However, performance drops in the imbalanced setting due to uneven class representation, highlighting the continued impact of data imbalance. This improvement suggests that Cross Attention effectively enhances feature representation by emphasizing clinically important retinal structures.

Table 4.8: Experimental results of balanced and imbalanced APTOS and IDRiD datasets with Cross Attention + EfficientNet-B4 backbone using Accuracy, F1-score, and QWK evaluation metrics.

Dataset	Method	Accuracy	F1 score	QWK
APTOS(Balanced)	CA + Backbone	0.8049	0.8014	0.8486
APTOS(Imbalanced)		0.7186	0.7116	0.8089
IDRiD (Balanced)		0.8185	0.8218	0.8886
IDRiD (Imbalanced)		0.5269	0.5318	0.7217

Table 4.9: Experimental results of balanced and imbalanced APTOS and IDRiD datasets with DenseNet-201 backbone using Accuracy, F1-score, and QWK evaluation metrics.

Dataset	Method	Accuracy	F1 score	QWK
APTOS(Balanced)	CA + Backbone	0.7678	0.7681	0.8328
APTOS(Imbalanced)		0.7134	0.6971	0.7649
IDRiD (Balanced)		0.9194	0.9178	0.9332
IDRiD (Imbalanced)		0.5678	0.6325	0.6945

4.6.3 Results with Region Attention and Backbone

The experimental results obtained using Region Attention (RA) with the EfficientNet-B4 backbone are presented in Table 4.10. For the APTOS dataset, the RA + Backbone model achieves performance close to the baseline but does not provide noticeable improvement. For the IDRiD dataset, Region Attention improves the balanced setting QWK score compared to the backbone baseline, suggesting better localization of lesion-specific regions. However, performance significantly decreased under the imbalanced setting, particularly in Accuracy and F1-score. The experimental results

obtained using Region Attention (RA) + DenseNet-201 backbone are presented in Table 4.11 for both balanced and imbalanced APTOS and IDRiD datasets. For the APTOS dataset, the balanced setting achieved higher Accuracy, F1-score, and QWK values than the imbalanced dataset, indicating that balanced class distribution allows the RA module to learn more discriminative regional features. Performance decreases under class imbalance due to biased representation of disease stages. On the IDRiD dataset, strong performance is observed in the balanced setting, demonstrating effective localization of pathological regions through Region Attention. However, performance drops significantly in the imbalanced scenario, highlighting that although RA improves feature learning, class imbalance still limits generalization. This behavior indicates that Region Attention alone is sensitive to data imbalance and may struggle to generalize when minority classes are underrepresented.

Table 4.10: Experimental results of balanced and imbalanced APTOS and IDRiD datasets with Region Attention + EfficientNet-B4 backbone using Accuracy, F1-score, and QWK evaluation metrics.

Dataset	Method	Accuracy	F1 score	QWK
APTOS(Balanced)	RA + Backbone	0.7926	0.7895	0.8470
APTOS(Imbalanced)		0.6995	0.6961	0.7649
IDRiD (Balanced)		0.7903	0.7872	0.8900
IDRiD (Imbalanced)		0.4462	0.3990	0.6797

Table 4.11: Experimental results of balanced and imbalanced APTOS and IDRiD datasets with DenseNet-201 backbone using Accuracy, F1-score, and QWK evaluation metrics.

Dataset	Method	Accuracy	F1 score	QWK
APTOS(Balanced)	RA + Backbone	0.8306	0.8290	0.8724
APTOS(Imbalanced)		0.7186	0.7116	0.8089
IDRiD (Balanced)		0.9355	0.9349	0.959
IDRiD (Imbalanced)		0.5269	0.5318	0.7217

4.6.4 Results with Multi-Stage Attention and Backbone

The experimental results of the Multi-Stage Attention Block (MSAB) with the EfficientNet-B4 backbone are presented in Table 4.12. The MSAB integrates both

Cross Attention (GATT) and Region Attention (SATT) block, enabling the model to learn global spatial information along with channel wise feature representation. Compared with the previous experiments, the MSAB + Backbone model achieves the best overall performance across both datasets. For APTOS dataset, the balanced setting shows improved Accuracy and QWK compared to the backbone-only, Cross Attention, and Region Attention models. For the IDRiD dataset, MSAB gives the state of the art results on balanced dataset as it achieves the highest Accuracy, F1-score, and QWK among all ablation configurations, confirming that combining global and regional attention enhances lesion discrimination. The experimental results obtained using the Multi-Stage Attention Block (MSAB) integrated with the DenseNet-201 backbone are presented in Table 4.13 for both balanced and imbalanced APTOS and IDRiD datasets. For the APTOS dataset, the balanced configuration achieved higher Accuracy, F1-score, and QWK compared to the imbalanced setting, demonstrating that multi-stage attention benefits from evenly distributed class samples. Performance slightly decreases in the imbalanced case due to unequal representation of disease severity levels. On the IDRiD dataset, the MSAB with DenseNet-201 achieved the best overall performance in the balanced setting, indicating effective feature refinement and enhanced lesion discrimination through multi-stage attention learning. However, performance drops under the imbalanced scenario, confirming that class imbalance still affects model generalization despite advanced attention integration. Even under the imbalanced setting, MSAB improved performance compared to Region Attention and backbone-only models, showing stronger generalization capability. Overall, the results confirm that the integration of both mechanisms in the proposed MSAB enables complementary feature learning, leading to more accurate and stable DR classification performance.

Table 4.12: Experimental results of balanced and imbalanced APTOS and IDRiD datasets with Multi-Stage Attention Block + EfficientNet-B4 backbone using Accuracy, F1-score, and QWK evaluation metrics.

Dataset	Method	Accuracy	F1 score	QWK
APTOS(Balanced)	MSAB + Backbone	0.8158	0.8053	0.8657
APTOS(Imbalanced)		0.7377	0.7276	0.8008
IDRiD (Balanced)		0.8226	0.8196	0.9021
IDRiD (Imbalanced)		0.5484	0.5535	0.6987

Table 4.13: Experimental results of balanced and imbalanced APTOS and IDRiD datasets with DenseNet-201 backbone using Accuracy, F1-score, and QWK evaluation metrics.

Dataset	Method	Accuracy	F1 score	QWK
APTOS(Balanced)	MSAB + Backbone	0.8142	0.8031	0.8481
APTOS(Imbalanced)		0.7186	0.7116	0.8089
IDRiD (Balanced)		0.9476	0.9469	0.9834
IDRiD (Imbalanced)		0.5269	0.5318	0.7217

4.7 Comparison with Existing Studies

Table 4.14 presents the comparison between the proposed framework and the base paper employing a Multi-Stage Attention Mechanism on the APTOS dataset. The base paper achieved an overall classification accuracy of 80%, with an F1-score of 0.79 and a Quadratic Weighted Kappa (QWK) score of 0.87, demonstrating reliable performance for diabetic retinopathy grading. In contrast, the proposed model based on EfficientNet-B4 integrated with the Multi-Stage Attention Block (MSAB) shows improved classification capability, achieving an accuracy of 81.58%. The proposed framework further enhances feature representation learning through refined attention mechanisms, enabling better discrimination among disease severity levels. Consequently, an improvement in F1-score (80.53) is observed, indicating stronger class-wise prediction consistency.

Similarly, Table 4.15 compares the proposed framework with several recent studies on DR classification. In recent years, researchers have explored different DL-based approaches for automated DR screening, including lightweight CNNs, hybrid architectures, lesion localization methods, and temporal modeling techniques. These approaches have significantly improved automated retinal image analysis and demonstrated the potential of AI-assisted DR diagnosis in clinical applications. To provide a more comprehensive evaluation, the proposed framework is tested on two publicly available datasets, namely APTOS and IDRiD. Both datasets contain retinal fundus images with different image characteristics, lesion distributions, and data complexities. The APTOS dataset includes images captured under real world conditions with noticeable variations in illumination, focus, and image quality, making classification more challenging. In contrast, the IDRiD dataset contains relatively

clearer and more standardized retinal images with well-defined lesion regions. Evaluating the model on both datasets helps analyze its ability to generalize across different imaging conditions and dataset characteristics. In addition, experiments are conducted under both balanced and imbalanced settings to further evaluate the stability of the proposed framework. Since DR datasets naturally suffer from class imbalance, minority classes often contain significantly fewer samples than majority classes. This imbalance can negatively affect model learning and bias predictions toward dominant classes. By evaluating the framework under both conditions, a more realistic understanding of the model's practical performance is achieved.

Earlier studies utilizing conventional CNN architectures [50] achieved an accuracy of 75.61%, while hybrid deep learning models improved performance to 79.50% [24], demonstrating the advantage of combining multiple feature learning strategies. Similarly, the multi-attention block [29] approach reported an accuracy of 80%, indicating that attention mechanisms significantly enhance lesion-focused feature extraction. Another lesion localization CNN [51] framework achieved 77.80% accuracy, highlighting the importance of spatial feature awareness in retinal image analysis. More advanced hybrid architectures combining VGG19 and ResNet50V2 [26] significantly increased performance to 91.8%, highlighting the benefit of deep feature fusion. Similarly, transformer-integrated CNN models [37] and Feature Extraction Block (FEB) with transformer approaches [39] achieved accuracy of 89.1% and 91.54%, respectively, demonstrating the effectiveness of attention mechanisms in medical image analysis.

The proposed EfficientNet-B4 with Multi-Stage Attention Block (MSAB) framework surpasses most existing methods under balanced dataset conditions. On the APTOS balanced dataset, the proposed model achieves an accuracy of 81.58%, outperforming traditional CNN, hybrid, and attention-based baseline approaches. Furthermore, evaluation on the IDRiD balanced dataset demonstrates even stronger performance with an accuracy of 82.26%, confirming the robustness and generalization capability of the proposed architecture across different datasets. However, performance decreases under imbalanced data conditions, where the proposed model achieves 73.77% on APTOS-ImBal and 54.84% on IDRiD-ImBal. This reduction highlights the inherent challenges posed by class imbalance in medical image classification tasks, particularly for minority disease severity classes. Furthermore, the proposed DenseNet-201 with MSAB

backbone achieved 81.42% accuracy on APTOS-Balanced, while significantly improving performance on the IDRiD-Balanced dataset with 94.76% accuracy, outperforming several previously reported CNN and hybrid approaches. These results confirm that dense feature reuse and enhanced information flow in DenseNet-201 effectively capture fine-grained retinal lesion characteristics.

Despite this limitation, the proposed framework maintains competitive performance compared to existing approaches and shows clear improvements when proper data balancing strategies are applied. Overall, the experimental results demonstrate that integrating EfficientNet-B4 and DenseNet-201 with a multi-stage attention mechanism improves feature representation, enhances lesion discrimination, and achieves superior classification performance compared to previously reported methods.

Table 4.14: Comparison of proposed framework with Base-Paper

Reference	Architecture	Dataset	Accuracy	F1	QWK
[29]	Multi-Stage attention Mechanism	APTOS	80%	0.79	0.87
Proposed	EffNet-B4+MSAB	APTOS	81.58%	80.53	86.57

Table 4.15: Comparison of proposed framework with Recent DR Classification Studies

Reference	Year	Architecture	Dataset	Accuracy
[50]	2022	CNN	APTOS	75.61%
[24]	2023	Hybrid		79.50%
[29]	2025	Multi-attention Block		80%
[51]	2025	Lesion Localization CNN		77.80%
[26]	2026	VGG19 + ResNet50V2 / Hybrid approach	APTOS/IDRiD	91.8%

[37]	2024	CNN, and Vision Transformer (ViT)		89.1%
[39]	2023	Feature Extraction Block (FEB) with transformer		91.54%
Proposed	EffNet-B4+MSAB		APTOS-Bal	81.58%
			APTOS-ImBal	73.77%
			IDRiD-Bal	82.26%
			IDRiD-ImBal	54.84%
Proposed	DenseNet-201+MSAB		APTOS-Bal	81.42%
			APTOS-ImBal	71.86%
			IDRiD-Bal	94.76%
			IDRiD-ImBal	52.69%

4.8 Discussion

This study presents a CNN-based framework for DR classification that integrates preprocessing and a Multi-Stage Attention mechanism to improve feature learning and classification performance. The experimental results demonstrate that the proposed approach effectively identifies different DR severity stages from retinal fundus images. The proposed framework achieves strong performance across multiple evaluation metrics, including accuracy, precision, recall, F1-score, and QWK. These results indicate the model's ability to distinguish between different DR stages with reliable classification capability. The preprocessing stage improves image quality and helps reduce the impact of class imbalance, while the attention mechanism enables the model to focus more effectively on clinically important retinal regions. This combination enhances feature representation and improves lesion-focused learning. The experimental findings of this study confirm that integrating an attention mechanism within a deep convolutional neural network significantly improves diabetic retinopathy classification performance. Traditional CNN-based approaches evaluated on the APTOS dataset achieved accuracies between 75.61% and 80%. In comparison, the

proposed EfficientNet-B4 with Multi-Stage Attention Block (MSAB) achieved a higher accuracy of 81.58% on the balanced APTOS dataset and 82.26% on the balanced IDRiD dataset [Answer to RQ1].

The integration of attention mechanisms significantly enhances deep learning models for diabetic retinopathy (DR) classification by addressing several real-world challenges. First, attention modules enable the network to automatically focus on clinically important retinal regions such as microaneurysms, hemorrhages, and exudates while suppressing irrelevant background information. This selective feature learning helps the model remain robust against image quality variations including blur, illumination changes, noise, and inconsistent acquisition conditions commonly present in real clinical datasets. Second, attention mechanisms help mitigate the class imbalance problem. In DR datasets, severe disease stages often have fewer samples compared to normal or mild classes. Attention enhances discriminative feature representation for minority classes by assigning higher weights to subtle pathological regions, improving sensitivity toward underrepresented DR grades. Finally, attention improves generalization across diverse datasets by capturing both local lesion details and global contextual relationships. By learning where and what to focus on, the model becomes less dependent on dataset-specific characteristics and performs consistently across different real-world datasets such as APTOS and IDRiD. As a result, the proposed attention-based framework achieves improved robustness, balanced learning, and stronger cross-dataset generalization performance [Answer to RQ2].

The proposed model achieved higher accuracy under balanced conditions, while performance decreased on imbalanced datasets (73.77% on APTOS-ImBal and 54.84% on IDRiD-ImBal). This observation highlights that although attention mechanisms enhance feature discrimination, class imbalance remains a challenging issue in medical imaging tasks. Nevertheless, the attention-guided architecture improves minority lesion detection compared to conventional CNN models by emphasizing subtle pathological patterns.

Chapter 5

Conclusion and Future Work

5.1 Conclusion

This study presented a deep learning-based framework for DR classification using retinal fundus images. The proposed framework integrates EfficientNet-B4 with a Multi-Stage Attention mechanism to improve feature extraction and classification performance. Preprocessing techniques were applied to enhance image quality and reduce the impact of class imbalance, while Cross-Patch Attention and Region Attention improved global contextual understanding and local lesion-focused feature refinement. The framework was evaluated on APTOS and IDRiD datasets under both balanced and imbalanced settings, where it achieved strong performance across multiple evaluation metrics, including accuracy, precision, recall, F1-score, and QWK. The ablation study further confirmed the contribution of each attention module toward improved feature learning and DR stage discrimination. Overall, the findings demonstrate that the proposed EfficientNet-B4 with Multi-Stage Attention framework provides an effective and reliable solution for DR classification by achieving a strong balance between classification performance, computational efficiency, and generalization capability, while also showing strong potential for practical retinal screening and clinical decision support applications.

5.2 Future Work

In future work, the proposed framework can be improved by using larger and more diverse retinal image datasets collected from multiple clinical centers to further enhance robustness and generalization capability. More advanced architectures, including transformer-based and hybrid attention models, can also be explored to improve global feature learning and classification performance. In addition, lightweight optimization techniques such as model pruning and quantization may help reduce computational complexity for deployment on low-resource medical devices and real time screening systems. Future research may also focus on integrating explainable AI techniques, including attention visualization and Grad-CAM, to improve interpretability and help clinicians better understand the model's decision-making process. Furthermore, the framework can be extended for multi-disease retinal analysis to support the detection of additional ocular diseases in practical clinical applications.

Chapter 6

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